

DEVELOPMENT OF BAYESIAN STATE-SPACE SURPLUS PRODUCTION MODEL JABBA FOR ASSESSING THE MEDITERRANEAN SWORDFISH (*XIPHIAS GLADIUS*) STOCK

Henning Winker¹; Ai Kimoto²; Bruno L. Mourato³; George Tserpes⁴ Mauricio Ortiz²

SUMMARY

*Bayesian State-Space Surplus Production Models were fitted to Mediterranean swordfish (*Xiphias gladius*) catch and CPUE data using the 'JABBA' R package. For the 'Reference' model we used a Schaefer model with an existing r prior, while for the 'ASEM' model we developed a Pella-Tomlinson model based on priors derived from an Age-structured Equilibrium Model. Both models produced reasonably robust model diagnostic results and predicted a similar exploitation history, with the peak in catches in 1982 (~ 20,000 tons) also demarcating the onset of an extended period of overfishing, which eventually led to an overfished stock. The stock status for 2018 showed that the 'ASEM' model was more optimistic, with a cumulative probability of over 60% (yellow + green) that current fishing mortality is currently sufficiently low enough to facilitate stock rebuilding. The 'Reference' model, by contrast, suggests that there is a more than 50% probability that stock remains overfished and that overfishing is still occurring. Sensitivity analysis suggests that these results dependent on the reliability of the relative magnitude of historical catches prior to 1987.*

RÉSUMÉ

*Les modèles de production excédentaire état-espace de type bayésien ont été ajustés aux données de capture et de CPUE de l'espadon de la Méditerranée (*Xiphias gladius*) au moyen du progiciel JABBA R. Pour le modèle de « référence », nous avons utilisé un modèle de Schaefer avec une distribution a priori existante de r , tandis que pour le modèle « ASEM », nous avons développé un modèle Pella-Tomlinson basé sur des distributions a priori dérivées d'un modèle en conditions d'équilibre structuré par âge. Les deux modèles ont produit des résultats diagnostiques raisonnablement robustes et ont prédit un historique d'exploitation similaire, le pic des captures en 1982 (~ 20.000 tonnes) délimitant également le début d'une période prolongée de surpêche, qui a finalement conduit à la surexploitation du stock. L'état du stock pour 2018 a montré que le modèle « ASEM » était plus optimiste, avec une probabilité cumulative de plus de 60 % (jaune + vert) que la mortalité par pêche actuelle soit suffisamment faible pour permettre le rétablissement du stock. Le modèle de « référence », en revanche, suggère qu'il y a plus de 50% de probabilité que le stock reste surpêché et que la surpêche se poursuive. L'analyse de sensibilité suggère que ces résultats dépendent de la fiabilité de l'ampleur relative des captures historiques antérieures à 1987.*

RESUMEN

*Los modelos de producción excedente bayesianos de estado espacio se ajustaron a los datos de captura y CPUE del pez espada del Mediterráneo (*Xiphias gladius*) utilizando el paquete R de «JABBA». Para el modelo de «referencia» se utilizó un modelo Schaefer con una distribución a priori para r existente, mientras que para el modelo «ASEM» se desarrolló un modelo Pella Tomlinson basado en distribuciones a priori derivadas de un modelo estructurado por edad en equilibrio. Ambos modelos produjeron resultados de diagnósticos razonablemente robustos y predijeron una historia de explotación similar, con el pico en las capturas en 1982 (~ 20,000 t) delimitando también el inicio de un amplio periodo de sobrepesca que, eventualmente, produjo*

¹ Joint Research Centre (JRC), European Commission, Ispra, Italy

² ICCAT Secretariat, Calle Corazón de María 8, Madrid Spain 28002.

³ Instituto do Mar, Universidade Federal de São Paulo, Av. Doutor Carvalho de Mendonça, 144, 11070-100, Santos, Brazil.

⁴ Hellenic Centre for Marine Research (HCMR), Greece

un stock sobrepescado. El estado del stock para 2018 mostraba que el modelo «ASEM» era más optimista, con una probabilidad acumulada de más del 60 % (amarillo+verde) de que la actual mortalidad por pesca es actualmente lo suficientemente baja para facilitar la recuperación del stock. El modelo de «referencia», por el contrario, sugería que existe más de un 50 % de probabilidad de que el stock permanezca sobrepescado y sufriendo sobrepescado y de que se esté produciendo sobrepesca. Los análisis de sensibilidad sugieren que estos resultados dependen de la fiabilidad de la magnitud relativa de las capturas históricos antes de 1987.

KEYWORDS

*Mediterranean Swordfish, stock status, CPUE fits, biomass dynamic model,
Pella-Tomlinson surplus production function*

1. Introduction

The broadbill swordfish (*Xiphias gladius*) is the most widely distributed billfish species. In the Atlantic Ocean and the Mediterranean Sea the species is intensively exploited. For assessments and management, the International Commission for the Conservation of Atlantic Tunas (ICCAT) considers three stocks in the North Atlantic, South Atlantic and the Mediterranean Sea (Neilson et al., 2013). The available catch histories of the three stocks are broadly comparable for much of the time both in terms of magnitude and trends. An exception is period prior 1980's in the North Atlantic, where the large development of swordfish directed longline fisheries by the USA, Canada, Portugal and Spain resulted in an earlier increase in catches than in the other regions (ICCAT, 2017a, 2017b) where catches started to increase later but then more steeply in 1980s. In the Mediterranean Sea there is however some evidence indicating that for instance in Italy catches prior to 1982 might have been substantially higher than reflected by the ICCAT catch time series (Piroddi et al., 2015). Similarly, it was noted that a relatively sharp increase in catches prior to the peak in total catches in 1988 (~20,000 t) may be partially attributed to improvements in the national systems for collecting catch statistics (ICCAT, 2017b). The surface long-line fishery developed slower in the Mediterranean than in the North Atlantic and a large proportion of swordfish were caught by gillnets throughout the 1990s and early 2000s (ICCAT, 2017b). Since 2012, reported gillnet catches have ceased and almost all swordfish landings are accrued to various longline fleets.

The most recent full stock assessment for Mediterranean swordfish was carried out by ICCAT in 2016. This assessment presented stock status estimates through 2014 from the Extended Survivor Analysis (XSA) using the 'FLXSA' in flr (Kell et al., 2007) and the non-equilibrium production model ASPIC (Prager, 2002), whereas the 2014 ICCAT assessment of Mediterranean swordfish (ICCAT, 2015) had also included a Bayesian Production Model (BSP; McAllister, 2014). Kolesidis and Tserpes (2020) presented a comparison between the two contemporary State-Space Surplus Production Models (SPMs), SPiCt (Pedersen and Berg, 2017) and JABBA (Winker et al., 2018), that were fitted to total catch data (1950-2018) and the standardized CPUE from Greek LL fleet. The reference case specification for assumed a Schaefer model with the same r prior that was developed during the 2014 assessment (ICCAT, 2015) and found that both SPiCt and JABBA provided consistent time trend estimates in relative biomass and relative fishing mortality with a rapid biomass decreases from the middle 1980s up to 2010, while the opposite trend was found for fishing mortality (Kolesidis and Tserpes, 2020). An important difference between previous applications of the age-structured XSA and the various SPM platforms (APIC, BSP, SPiCt, JABBA) to Mediterranean swordfish is that all SPMs were fitted with the entire catch time series, starting in 1950, while XSA was only initiated in 1987. As a result, the SPMs consistently predicted that biomass had declined to levels around or just below B_{MSY} by 1987, whereas XSA that was initiated in 1987 and predicted that the stock was already collapsed by that time with stock biomass below half of B_{MSY} . Although, the 2016 ICCAT assessment were fairly conclusive in that the stock biomass was overfished and overfishing was occurring, there were substantial differences between the models in terms of the degree of overfishing and optimal harvest levels (ICCAT, 2017b).

Here we present stock assessment results for Mediterranean swordfish stock based on the Bayesian State-Space Surplus Production Model framework, JABBA (Just Another Bayesian Biomass Assessment; Winker et al., 2018a), using updated catch and standardized longline CPUE time series through 2017. JABBA is a fully documented, open-source R package ([www.github.com/JABBAmodel](https://github.com/JABBAmodel)) that has been formally included in the ICCAT stock catalogue (<https://github.com/ICCAT/software/wiki/2.8-JABBA>). JABBA has been applied in a number of recent ICCAT stock assessments, including south Atlantic blue shark (ICCAT, 2016), Mediterranean

albacore (ICCAT, 2017c), south Atlantic swordfish (ICCAT, 2017a; Winker et al., 2018), Atlantic shortfin mako shark stocks (south and north) (ICCAT, 2017d; Winker et al., 2017, 2019a), Atlantic blue marlin (Mourato et al., 2019), Atlantic bigeye tuna (Winker et al., 2019b), Atlantic White marlin (Mourato et al., 2020) and Atlantic yellowfin tuna (Sant'Ana et al., 2020).

2. Material and Methods

This preliminary stock assessment is implemented using the Bayesian state-space surplus production model framework JABBA (Winker et al., 2018a), which is now available as ‘R package’ that can be installed from github.com/jabbamodel/JABBA. JABBA’s inbuilt options include: (1) automatic fitting of multiple CPUE time series and associated standard errors; (2) estimating or fixing the process variance, (3) optional estimation of additional observation variance for individual or grouped CPUE time series, and (4) specifying a Fox, Schaefer or Pella-Tomlinson production function by setting the inflection point B_{MSY}/K and converting this ratio into shape a parameter m , (5) a number diagnostic (e.g. residual run tests) and (6) a routine to conduct retrospective analysis.

A full JABBA model description, including formulation and state-space implementation, prior specification options and diagnostic tools is available in Winker et al (2018a).

2.1. Fishery data

Fishery catch data for Mediterranean swordfish were available by ICCAT secretariat for the period 1950-2018 (**Figure 1**). Relative abundance indices were made available in the form of standardized catch-per-unit-of-effort (CPUE) time series. The standardized CPUE series cover a variety time periods of fishing fleets, including longline (LL), drift gillnet (Gil) fisheries. Here, we only considered four standardized LL CPUE indices from Greece (GrLL; 1987 - 2018), Spain (SpLL; 1988-2018), Liguria (LiLL; 2010-2018) in Italy; and Morocco (MoLL; 2012-2018). All four standardized CPUE indices analyzed for both catch rates in numbers and in weight. For fitting the surplus production model JABBA, we only used CPUE in weight (**Figure 2**) and assumed those to be proportional to biomass.

2.2. Model specifications

For the unfished equilibrium biomass K , we used default settings of the JABBA R package in the form of vaguely informative lognormal prior with a large CV of 100% and a central value that corresponds to eight times the maximum total catch and is consistent with other platforms such as Catch-MSY (Martell and Froese, 2013) or SpiCt (Pederson and Berg 2017). Initial depletion lognormal prior ($\phi = B_{1950}/K$) was inputted with mean = 1 and CV of 10%, consistent with the understanding that there was very little fishing before the starting year of 1950 (Kolesidis and Tserpes, 2020). All catchability parameters were formulated as uninformative uniform priors, while additional observation variances were estimated for index by assuming inverse-gamma priors to enable model internal variance weighting. Instead, the process error of $\log(B_y)$ in year y was estimated “freely” by the model using an uninformative inverse-gamma distribution with both scaling parameters setting at 0.001.

For the ‘Reference’ model we assumed a prior distribution for r with mean of 0.47 and CV of 0.49 ($\log.sd = 0.46$) which was derived from a Leslie matrix model approach with Monte-Carlo simulations during the 2014 Mediterranean Swordfish Stock Assessment meeting (ICCAT, 2015). This approach combines estimates of von Bertalanffy growth, weight/length relationship, logistic maturity ogive, natural mortality and the steepness parameter of the spawner recruitment relation and r can then be calculated as the dominant eigenvalue from a Leslie matrix (McAllister et al., 2000; Thorson, 2019). In agreement with the 2016 assessment and the recent work by Kolesidis and Tserpes (2020), the shape of the production function was described by Schaefer model, so that MSY is attained at $B_{MSY}/K = 0.5$.

In addition, we developed an alternative r prior distribution with an associated shape parameter of a Pella-Tomlinson production function from an Age-Structured Equilibrium Model (ASEM) approach with Monte-Carlo simulations (Chang et al., 2019; Winker et al., 2020b, 2020a). Central to the concept approach of linking age-structured stock parameters and the surplus production function is the application of age-structured spawning biomass- and yield-per-recruit models with integrated stock-recruitment relationship (SRR), which we subsequently refer to as ASEM. This type of model is widely used in age-structured to derive MSY-based fisheries reference points (FRPs) from estimated stock parameters by searching iteratively for the fishing mortality that produces MSY, F_{MSY} , from the corresponding biomass B_{MSY} at MSY. The ASEM stock parameters

are conceptually the same as for the Leslie-Matrix approach (**Table 1**). However, there are two key differences between the ASEM approach and the Leslie matrix approach: the ASEM requires (1) the additional input of selectivity a function to account for the impact of selectivity on surplus production curve of the exploitable biomass (EB), and (2) the shape parameter m is directly derived from the ASEM output of EB_{MSY}/EB_0 (Winker et al., 2020b).

The stock parameters used as input for the ASEM included the following three modifications compared to the 2014 assessment (ICCAT, 2015): (i) revised parameters of the length-weight relation, (ii) a revised estimate of length-at-50%-maturity from previously 142 cm to 132 cm and (iii) an increased maximum from 8 to 10 years. In addition, to a constant $M = 0.206$, we also explored an age-dependent vector M_a estimates based on the weight-based equation by Lorenzen (1996), which was tuned so that $M_{at-3} = 0.2$ to be consistent with the corresponding 2016 XSA assessment assumption (**Figure 3**). Size-at-50%-selectivity was set to 70 cm, which results in 100% of age-1 being vulnerable to the fishery and about 50% age-0 fish. To account for uncertainty about these choices we generated these parameters from lognormal distributions with CVs ranging from 15% for length-at-50%-maturity to large CVs of 35% for natural mortality.

Despite admitting considerable uncertainty about key stock parameters, the r prior distributions were relatively precise ($\sim CV = 20\%$) and showed little contrast between the constant and the age-dependent M assumptions with central values of around $r = 0.2$ and a shape parameter corresponding to B_{MSY}/K (**Figure 4**; **Table 2**). As a result we combined the r prior distributions generated from the two M assumptions into a single joint r prior (**Table 2**).

Accordingly, we formulated the following two model scenarios for Mediterranean swordfish

- **Reference (2014):**, with $\log(r) \sim N(\log(0.45), 0.46)$ and a fixed input value of $B_{MSY}/K = 0.5$ (Schaefer)
- **ASEM (2020):** with $\log(r) \sim N(\log(0.206), 0.2)$ and a fixed input value of $B_{MSY}/K = 0.38$

2.3. Model diagnostics and sensitivity runs

JABBA is implemented in R (R Development Core Team, <https://www.r-project.org/>) with JAGS interface (Plummer, 2003) to estimate the Bayesian posterior distributions of all quantities of interest by means of a Markov Chains Monte Carlo (MCMC) simulation. The JAGS model is executed from R using the wrapper function `jags()` from the library `r2jags` (Su and Yajima, 2012), which depends on `rjags`. In this study, two MCMC chains were used. Each model was run for 30,000 iterations, sampled with a burn-in period of 5,000 for each chain and thinning rate of five iterations. Basic diagnostics of model convergence included visualization of the MCMC chains using MCMC trace-plots as well as Heidelberger and Welch (Heidelberger and Welch, 1992) and Geweke (1992) and Gelman and Rubin (1992) diagnostics as implemented in the coda package (Plummer et al., 2006).

To evaluate CPUE fits, the model predicted CPUE indices were compared to the observed CPUE. JABBA-residual plots were used to examine (1) colour-coded lognormal residuals of observed versus predicted CPUE indices by fleet together with (2) boxplots indicating the median and quantiles of all residuals available for any given year; the area of each box indicates the strength of the discrepancy between CPUE series (larger box means higher degree of conflicting information) and (3) a loess smoother through all residuals which highlights systematically auto-correlated residual patterns to evaluate the randomness of model residuals. In addition it depicts the root-mean-squared-error (RMSE) as a goodness-of-fit statistic. We conducted a run test to quantitatively evaluate the randomness of residuals (Carvalho et al., 2017). The run test diagnostic was applied to residuals of the CPUE fit on log-scale using the function `runs.test` in the R package `tseries`, considering the 2-sided p-value of the Wald-Wolfowitz runs test. The runs test results can be visualized within JABBA using a specifically designed plot function that illustrates which time series passed or failed the runs test and highlights individual time-series data points fall outside the three-sigma limits (e.g. Anhøj and Olesen, 2014).

To examine the sensitivity of the assessment results to the inclusion of individual CPUE indices, we iteratively re-fitted the models while excluding one of the four indices at the time (Jackknife index analysis). In addition, we explored the robustness to various levels of under reporting prior to 1987. To do this we created four alternative catch scenarios by multiplying the early catches from 1950 to 1986 by factors of 1.2, 1.5, 2 and 3, (i.e underreporting of 20%, 50%, 100% and 200%) and fitted JABBA to each of the ‘new’ catch series with otherwise unchanged settings (**Figure 5**).

To check for systematic bias in the stock status estimates, we also performed a retrospective analysis for each model, by removing one year of data at a time sequentially ($n = 5$), refitting the model and comparing quantities of interest (i.e. biomass, fishing mortality, B/B_{MSY} , F/F_{MSY} , B/B_0 and MSY) to the reference model that is fitted to full time series. To compare the bias between the models, we computed Mohn's (1999) rho (ρ) statistic and specifically the commonly used formulation Hurtado-Ferro et al. (2014).

3. Results and Discussion

The MCMC convergence tests by Heidelberger and Welch (Heidelberger and Welch, 1992) and Geweke (1992) and Gelman and Rubin (1992) were passed by all estimable key parameters for both the 'Reference' and the 'ASEM' models. Adequate convergence of the MCMC chains was also corroborated by visual inspection of trace plots (results available on request), which showed good mixing in general (i.e., moving around the parameter space).

The model fit to each of the four standardized CPUE LL indices are shown in **Figures 6** and **7**. Both models appeared to fit CPUE data reasonably well, and run tests conducted on the log-residuals provided no evidence to reject the hypothesis of randomly distributed residual patterns for all four indices (**Figures 8** and **9**). Generally, the goodness-of-fit was very similar for the 'Reference' model (RMSE = 28.5%; **Figures 10**) and the 'ASEM' model (RMSE = 29%; **Figure 10**). The residual patterns of the last two years 2017-2018 indicated a conflict between positive residuals for Spanish LL CPUE fits and negative residuals for the other three CPUE indices (**Figure 10**). This resulted in an, on average, negative trend in the residual pattern in most recent years as shown in the JABBA-residual plots. The estimated decreasing trend in abundance in most recent years (**Figures 6** and **7**) coincides with decrease in landing. As a result, the estimated process error deviates show a negative trend in most recent years, which also means that both models predict that the stock's productivity has been below average in recent years. This is consistent with 2016 XSA results, which indicated that recruitment had been below its deterministic expectation over an extended period (ICCAT, 2017b).

Marginal posterior distributions along with prior densities are shown in **Figures 11** and **12**. The medians of marginal posteriors for r were 0.414 and 0.188 for the 'Reference' and the 'ASEM' models, respectively (**Table 3**). The small prior to posterior median ratio (PPMR) for r indicated a good agreement and overlapped well between the density distributions. The smaller prior to posterior variance ratio (PPVR) of r for the 'Reference' can be ascribed to the relatively larger CV. However, it also indicates that the data are to some extent informative about r . The estimated median of marginal posterior for K was 135,017 metric tons for the 'Reference' model and 194,523 metric tons for the 'ASEM' model. It was also noted that posterior distributions for K were narrower in comparison to their priors, The resulting small PPVRs indicate that the input data was very informative about K for two models (**Figures 11** and **12**). The marginal posteriors for initial depletion (ϕ) were similar between models, with both PPMR and PPVR close to 1, which suggests that the marginal posteriors of both models were largely informed by the priors.

Summaries of posterior quantiles for parameters and management quantities of interest are presented in **Table 3**. Estimates of MSY showed similar values between models (13,811 metric tons for the 'Reference' model and 12,931 metric tons for the 'ASEM' model, **Table 3**). The marginal posterior median for B_{MSY} varied between 66,513 metric tons ('Reference' model) and 74,508 metric tons ('ASEM' model) (**Table 3** and **Figures 11** and **12**). The F_{MSY} median estimates were slightly higher for the 'Reference' model (0.207) than the 'ASEM' model (0.176) (**Table 3**). In general, both models showed similar trends for the medians of B/B_{MSY} and F/F_{MSY} over time, and the discrepancy of values between the models became less since the 1990s (**Figure 13**). The trajectory of B/B_{MSY} showed an overall decreasing trend from 1970 to the most recent year. The decreasing in the estimated biomass trend was steeper between 1970 and 1990 compared to the one after 1990. The estimated biomass has been less than the MSY level since the mid-1990s for the 'Reference' model ($B_{2018}/B_{MSY} = 0.650$), whereas it has dropped under the MSY level since the mid-2010s for the 'ASEM' model ($B_{2018}/B_{MSY} = 0.777$). The F/F_{MSY} trajectory showed a gradual increasing trend between 1970 and the mid-1980s, a sharp increase in the late-1980s (around 0.5 to 1.0 in a few years), and the relatively stable afterwards (**Figure 13**). Since the late 1980s after the sharp increase, the estimated fishing mortality have been fluctuated at over MSY level for the 'Reference' model ($F_{2018}/F_{MSY}=0.990$) and at around MSY level for the 'ASEM' model ($F_{2018}/F_{MSY}=0.880$) (**Table 3**).

A retrospective analysis for five years was also presented in **Figures 15** and **16**. **Table 4** depicts the Mohn's rho statistic computed for a retrospective evaluation period of five years for both models. The estimated Mohn's rho for B and B/B_{MSY} fell within the acceptable range of -0.15 and 0.20 (Hurtado-Ferro et al. 2014; Carvalho et al. 2017) and consequently indicated that the retrospective pattern was relatively small for both models. However, the 'Reference' model indicated slightly stronger retrospective patterns with regards to stock status trajectories B/B_{MSY} , F/F_{MSY} and B/B_0 for the retrospective runs through in 2013 and 2014, which may also explain the notably smaller retrospective bias for these quantities in the 'ASEM' model.

The Jackknife sensitivity analysis of CPUE indices showed that Greek and the Spanish LL CPUE were highly influential with regards to stock status trajectories and MSY (**Figures 16** and **17**). Removing the Greek LL index resulted in much more optimistic stock status trajectories, with biomass level well above B_{MSY} . Removing the Spanish LL index, would result in considerably more pessimistic stock trajectories, estimating that the stock was overfished since the mid-1990s. The short CPUE time series from Liguria and Morocco appear to corroborate the stronger decreasing trend in the Greek LL in most recent years and exclusion of any of the indices would therefore result in more optimistic stock status (**Figures 16** and **17**). The 'ASEM' model was generally slightly less sensitive to the Jackknife index analysis (**Figure 16**) when compared to the 'Reference' model (**Figure 17**).

The second sensitivity analysis explored the robustness to various levels of under reporting prior to 1987 by multiplying the early catches from 1950 to 1986 by factors of 1.2, 1.5, 2 and 3, (i.e. assumed underreporting of 20%, 50%, 100% and 200%, **Figure 5**). Contrary to the Jackknife sensitivity analysis, the catch scenarios with the factors 1.2 and 1.5 affected the stock trajectories of the 'Reference' model less (**Figure 18**) than the 'ASEM' model (**Figure 19**). For both models the increasingly raised historical catch time series resulted in a systematic increase in the estimates of k and MSY (**Figure 18** and **19**). At the same time, the decline in stock biomass between 1970 and the late 1980s became more pronounced. Raising the historical catch by a factor three would result a decrease in biomass levels of about 50% of B/B_{MSY} by 1990, which be broadly similar to the results from the 2016 XSA assessment model results (ICCAT, 2017b).

JABBA surplus production phase plots for the 'Reference' and 'ASEM' models are shown in **Figure 20**. In general, it revealed similar patterns for both models. The peak in catches in 1982 concurred with the onset of an extended period of overfishing associated with continues declines in stock biomass and eventually led to an overfished stock. Accordingly, Kobe biplots show the typically anti-clockwise pattern with the stock status moving from underexploited through a period of unsustainable fishing to the overexploited phase (**Figure 20**). The resulting stock status posteriors for 2018 from each model would however have somewhat different implications for management actions. The 'ASEM' model is more optimistic with regards the current fishing levels, F_{2018}/F_{MSY} , with a cumulative probability of over 60% (yellow + green) that current fishing mortality is sufficiently low enough to facilitate stock rebuilding. The 'Reference', by contrast, suggests that there is a more than 50% probably that stock remains overfish and that overfishing is still occurring (**Figure 20**).

In conclusion, our results suggest that both candidate models provide reasonably robust fits to the data as judged by the presented model diagnostic results. Both models performed very similar with regards to the residuals diagnostics. The most notable difference in terms of performance is the slightly improved retrospective pattern for the 'ASEM' model. However, perhaps more importantly then the choice of candidate model is the importance of using reliable historical catch magnitudes relative to recent catches as input into for JABBA assessment models.

References

- Anhøj, J., Olesen, A.V., 2014. Run charts revisited: A simulation study of run chart rules for detection of non-random variation in health care processes. *PLoS One* 9, 1–13. <https://doi.org/10.1371/journal.pone.0113825>
- Carvalho, F., Punt, A.E., Chang, Y.J., Maunder, M.N., Piner, K.R., 2017. Can diagnostic tests help identify model misspecification in integrated stock assessments? *Fish. Res.* 192, 28–40. <https://doi.org/10.1016/j.fishres.2016.09.018>
- Chang, Y.J., Winker, H., Sculley, M., Hsu, J., 2019. Evaluation of the status and risk of overexploitation of the Pacific billfish stocks considering non-stationary population processes. *Deep. Res. Part II Top. Stud. Oceanogr.* in press. <https://doi.org/10.1016/j.dsr2.2019.104707>
- Gelman, A., Rubin, D.B., 1992. Inference from Iterative Simulation Using Multiple Sequences. *Stat. Sci.* 7, 457–472. <https://doi.org/10.2307/2246093>
- Geweke, J., 1992. Evaluating the accuracy of sampling-based approaches to the calculation of posterior moments., in: Berger, J.O., Bernardo, J.M., Dawid, A.P., Smith, A.F.M. (Eds.), *Bayesian Statistics 4: Proceedings of the Fourth Valencia International Meeting*. Clarendon Press, Oxford, pp. 169–193.
- Heidelberger, P., Welch, P.D., 1992. Simulation run length control in the presence of an initial transient. *Oper. Res.* 31, 1109–1144. <https://doi.org/10.1287/opre.31.6.1109>
- Hurtado-Ferro, F., Szuwalski, C.S., Valero, J.L., Anderson, S.C., Cunningham, C.J., Johnson, K.F., Licandeo, R., McGilliard, C.R., Monnahan, C.C., Muradian, M.L., Ono, K., Vert-Pre, K.A., Whitten, A.R., Punt, A.E., 2014. Looking in the rear-view mirror: Bias and retrospective patterns in integrated, age-structured stock assessment models, in: *ICES Journal of Marine Science*. pp. 99–110. <https://doi.org/10.1093/icesjms/fsu198>
- ICCAT, 2017a. Report of the 2017 ICCAT Atlantic swordfish stock assessment session. *Collect. Vol. Sci. Pap. ICCAT* 74, 841–967.
- ICCAT, 2017b. Report of the 2016 Mediterranean swordfish stock assessment meeting. *Col. Vol. Sci. Pap. ICCAT* 73, 1005–1096.
- ICCAT, 2017c. Report of the 2017 ICCAT albacore species group intersessional meeting (including assessment of Mediterranean albacore). *Collect. Vol. Sci. Pap. ICCAT* 74, 45.
- ICCAT, 2017d. Report of the 2017 ICCAT shortfin mako assessment meeting. *Collect. Vol. Sci. Pap. ICCAT* 74, 1465–1561.
- ICCAT, 2016. Report of the 2015 ICCAT blue shark stock assessment session. *Collect. Vol. Sci. Pap. ICCAT* 72, 866–1019.
- ICCAT, 2015. Report of the 2014 Mediterranean swordfish stock assessment meeting. *Col. Vol. Sci. Pap. ICCAT* 71, 1870–1979.
- Kell, L.T., Mosqueira, I., Grosjean, P., Fromentin, J., Garcia, D., Hillary, R., Jardim, E., Mardle, S., Pastoors, M.A., Poos, J.J., Scott, F., Scott, R.D., 2007. FLR: an open-source framework for the evaluation and development of management strategies. *ICES J. Mar. Sci.* 64, 640–646.
- Kolesidis, D.A., Tserpes, G., 2020. Preliminary assessment of the Mediterranean swordfish stock by means of Bayesian Surplus Production Models. *Col. Vol. Sci. Pap. ICCAT* 76, 193–204.
- Martell, S., Froese, R., 2013. A simple method for estimating MSY from catch and resilience 504–514. <https://doi.org/10.1111/j.1467-2979.2012.00485.x>
- McAllister, M.K., 2014. A generalized Bayesian surplus production stock assessment software (BSP2). *Collect. Vol. Sci. Pap. -ICCAT* 70, 1725–1757.

- McAllister, M.K., Babcock, E.A., Pikitch, E.K., Prager, M.H., 2000. Application of a non-equilibrium generalized production model to South and North Atlantic swordfish: combining Bayesian and demographic methods for parameter estimation. *Collect. Vol. Sci. Pap. ICCAT* 51, 1523–1550.
- Mohn, R., 1999. The retrospective problem in sequential population analysis: An investigation using cod fishery and simulated data. *ICES J. Mar. Sci.* 56, 473–488. <https://doi.org/10.1006/jmsc.1999.0481>
- Mourato, B.L., Winker, H., Carvalho, F., Kimoto, A., Ortiz, M., 2020. Developing of Bayesian State-Space Surplus Production JABBA for Assessing Atlantic white marlin (*Kajikia albida*) stock. *Col. Vol. Sci. Pap. ICCAT* 76, 235–254.
- Mourato, B.L., Winker, H., Carvalho, F., Ortiz, M., 2019. Stock Assessment of blue marlin (*Makaira nigricans*) using a Bayesian State-Space Surplus Production Model JABBA. *Collect. Vol. Sci. Pap. ICCAT* 75, 1003–1025.
- Neilson, J., Arocha, F., Cass-Calay, S., Mejuto, J., Ortiz, M., Scott, G., Smith, C., Travassos, P., Tserpes, G., Andrushchenko, I., 2013. The Recovery of Atlantic Swordfish: The Comparative Roles of the Regional Fisheries Management Organization and Species Biology. *Rev. Fish. Sci.* 21, 59–97. <https://doi.org/10.1080/10641262.2012.754842>
- Pedersen, M.W., Berg, C.W., 2017. A stochastic surplus production model in continuous time. *Fish Fish.* 18, 226–243. <https://doi.org/10.1111/faf.12174>
- Piroddi, C., Gristina, M., Zylich, K., Greer, K., Ulman, A., Zeller, D., Pauly, D., 2015. Reconstruction of Italy's marine fisheries removals and fishing capacity, 1950–2010. *Fish. Res.* 172, 137–147. <https://doi.org/10.1016/j.fishres.2015.06.028>
- Plummer, M., 2003. JAGS: A Program for Analysis of Bayesian Graphical Models using Gibbs Sampling, 3rd International Workshop on Distributed Statistical Computing (DSC 2003); Vienna, Austria.
- Plummer, M., Nicky Best, Cowles, K., Vines, K., 2006. CODA: Convergence Diagnosis and Output Analysis for MCMC. *R News* 6, 7–11.
- Prager, M.H., 2002. Comparison of logistic and generalized surplus-production models applied to swordfish, *Xiphias gladius*, in the north Atlantic Ocean. *Fish. Res.* 58, 41–57. [https://doi.org/10.1016/S0165-7836\(01\)00358-7](https://doi.org/10.1016/S0165-7836(01)00358-7)
- Sant'Ana, R., Mourato, B., Kimoto, A., Walter, J., Winker, H., 2020. Atlantic Yellowfin tuna stock assessment: An Implementation of a Bayesian State-Space Surplus Production Model using JABBA. *Col. Vol. Sci. Pap. ICCAT* 76, 699–724.
- Thorson, J.T., 2019. Predicting recruitment density dependence and intrinsic growth rate for all fishes worldwide using a data-integrated life-history model. *Fish Fish.* n/a. <https://doi.org/10.1111/faf.12427>
- Winker, H., Carvalho, F., Kapur, M., 2018. JABBA: Just Another Bayesian Biomass Assessment. *Fish. Res.* 204, 275–288. <https://doi.org/http://doi.org/10.1016/j.fishres.2018.03.01>
- Winker, H., Carvalho, F., Kerwath, S., 2019a. Age-structured biomass dynamics of north Atlantic shortfin mako with implications for the interpretation of surplus production models. *ICCAT-SCRS* 098, 1–19.
- Winker, H., Carvalho, F., Sharma, R., Parker, D., Kerwath, S., 2017. Initial results for North and South Atlantic shortfin mako (*Isurus oxyrinchus*) stock assessments using the Bayesian surplus production model JABBA and the catch-resilience method CMSY 74, 1836–1866.

- Winker, H., Carvalho, F., Thorson, J.T., Kell, L.T., Parker, D., Kapur, M., Sharma, R., Booth, A.J., Kerwath, S.E., 2020a. JABBA-Select: Incorporating life history and fisheries' selectivity into surplus production models. *Fish. Res.* 222, 105355. <https://doi.org/https://doi.org/10.1016/j.fishres.2019.105355>
- Winker, H., Kerwath, S.E., Merino, G., Ortiz, M., 2019b. Bayesian State-Space Surplus Production Model JABBA of Atlantic bigeye tuna (*Thunnus obesus*) stock. *Col. Vol. Sci. Pap. ICCAT* 75, 2129–2168.
- Winker, H., Mourato, B., Chang, Y., 2020b. Unifying parameterizations between age-structured and surplus production models: An application to Atlantic white marlin (*Kajika albida*) with simulation testing. *Col. Vol. Sci. Pap. ICCAT* 76, 219–234.

Table 1. Life history parameters for Mediterranean swordfish used to estimate r prior distributions and median shape parameter with corresponding B_{MSY}/K values generated an Age-Structured Equilibrium Model (ASEM) with a Beverton-Holt Stock-Recruitment relationship using Monte-Carlo simulations

Parameter	Mean	CV	Distribution	Description	Source
r	0.206*	0.35	Lognormal	Natural Mortality (1/year)	McAllister (2014)
L_{inf} (cm)	238.58	0.1	Lognormal	Von Bertalanffy asymptotic length	Mean: ICCAT Manual; CV: Working Group
K	0.185	0.1	Normal	Von Bertalanffy growth parameter	Mean: ICCAT Manual; CV: Working Group
t_0	-1.404	0.2	Normal	Von Bertalanffy age at zero length	Mean: ICCAT Manual; CV: Working Group
A	9.61E-06	-	Exponential	Weight at length parameter (GG-LJFL)	Tserpes et al. (2017)
B	3.059	-	Exponential	Weight at length parameter (GG-LJFL)	Tserpes et al. (2017)
L_{50} (cm)	132	0.2	Lognormal	Length at 50% maturity	Mean: ICCAT Manual; CV: McAllister (2014)
D	0.2	0.2	Lognormal	Parameter of the logistic maturity ogive	Working Group
t_{max} (years)	10	0.2	Lognormal	Longevity	Intersessionally
L_c (cm)	70	fixed	Fixed	Length at 50% selectivity	Intersessionally
H	0.83	fixed	Fixed	Steepness	McAllister (2014)

* In addition an age-dependent M_a was derived based on Lorenzen (1996) and tuned so that M -at-age-3 = 0.2.

Table 2. Results for r prior distributions and median shape parameter with corresponding B_{MSY}/K values generated an Age-Structured Equilibrium Model (ASEM) with a Beverton-Holt Stock-Recruitment relationship using Monte-Carlo simulations

Parameter	Scenario			
	Reference	M Constant	M Lorenzen	Joint
r	0.49	0.186	0.176	0.181
sd of $\log(r)$	0.47	0.198	0.207	0.204
B_{MSY}/K	0.5	0.38	0.38	0.38
shape m	2	1.06	1.08	1.07

Table 3. Summary of posterior quantiles presented in the form of marginal posterior medians and associated the 95% credibility intervals of parameters for the Bayesian state-space surplus production models for Mediterranean swordfish.

Estimates	<i>Reference model</i>			<i>ASEM model</i>		
	Median	2.50%	97.50%	Median	2.50%	97.50%
K	135017	79495	259387	194523	139243	290538
r	0.414	0.219	0.748	0.188	0.129	0.268
ψ (psi)	0.996	0.821	1.204	0.994	0.820	1.200
σ_{proc}	0.094	0.045	0.163	0.084	0.038	0.150
F_{MSY}	0.207	0.11	0.374	0.176	0.121	0.251
B_{MSY}	67509	39748	129693	73928	52919	110417
MSY	13811	11275	21476	12931	10709	17468
B_{1950}/K	0.994	0.75	1.267	0.989	0.758	1.254
B_{2018}/K	0.325	0.188	0.665	0.295	0.153	0.536
B_{2018}/B_{MSY}	0.650	0.376	1.330	0.777	0.404	1.41
F_{2018}/F_{MSY}	0.990	0.312	1.686	0.880	0.373	1.679

Table 4. Summary Mohn's rho statistic computed for a retrospective evaluation period of five years. The larger the threshold the stronger is the retrospective bias.

Scenario	Stock Quantity					
	B	F	B/B_{MSY}	F/F_{MSY}	B/K	MSY
Reference	0.061	-0.057	0.158	-0.173	0.158	0.045
ASEM	0.086	-0.078	0.101	-0.132	0.101	0.042

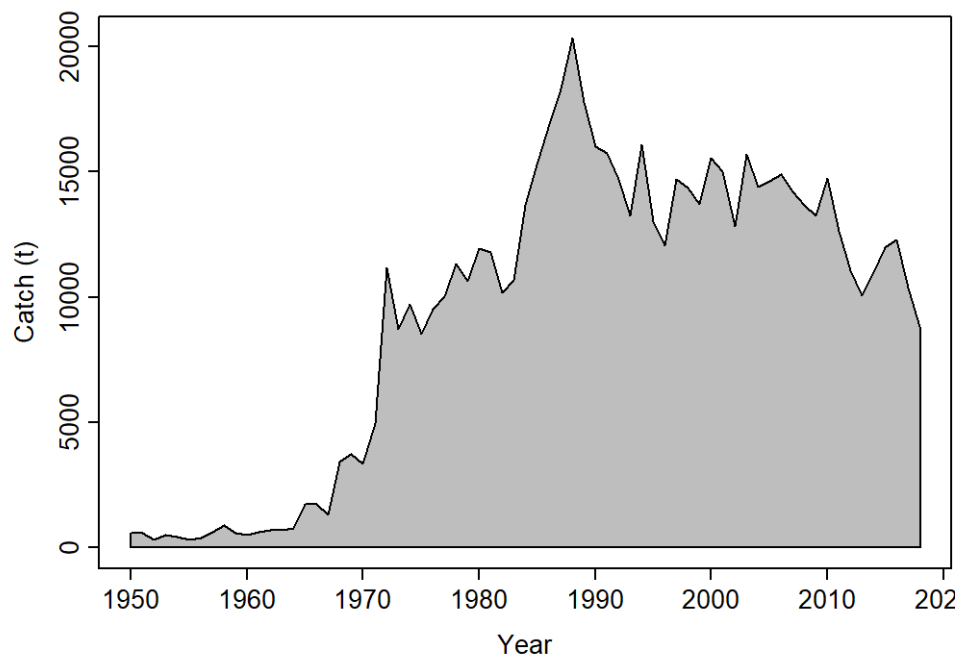


Figure 1. Available catch times series in metric tons (t) for Mediterranean swordfish.

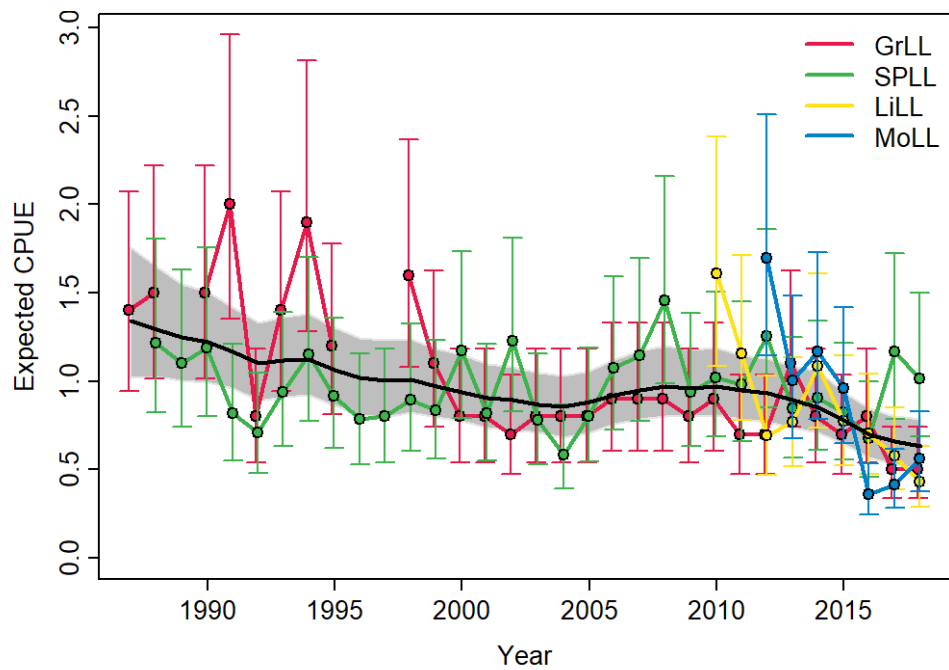


Figure 2. Time-series of four standardized CPUE series for Mediterranean swordfish with and assumed Standard Errors of 0.2. Solid black line is the averaged CPUE with associated confidence intervals (shaded area).

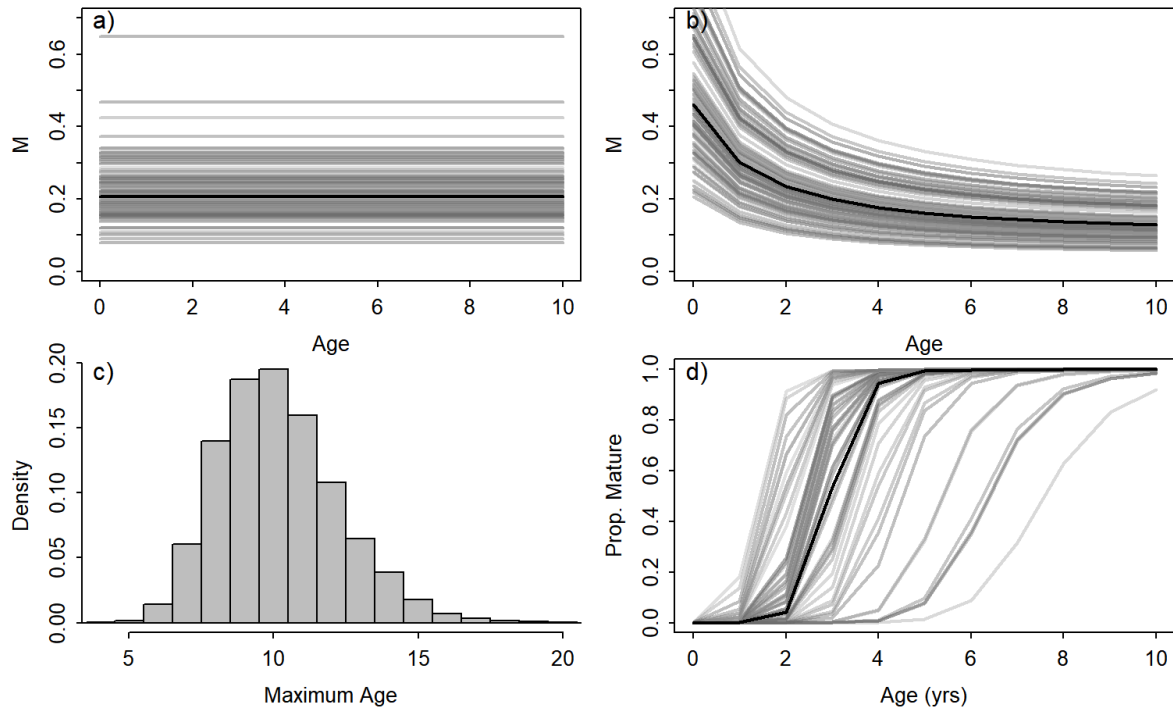


Figure 3. Admitted uncertainty about key stock parameters for Mediterranean swordfish used to estimate r prior distributions and median shape parameter with corresponding B_{MSY}/K values generated an Age-Structured Equilibrium Model (ASEM) with a Beverton-Holt Stock-Recruitment relationship ($h = 0.83$) using Monte-Carlo simulations (a) estimates of constant $M = 0.2$ with a CV = 35%; (b) estimates of alternative age-dependent with a CV = 35%, (c) histogram of maximum age integers generated from a lognormal with a mean of 10 years and a CV = 20% and (d) proportion mature females at age for $L_{50} = 132$ cm LJFL with a CV = 20%; (e) selectivity-at-length and; (f) specific selectivity-at-age as a function of specific length-at-age and selectivity-at-length.

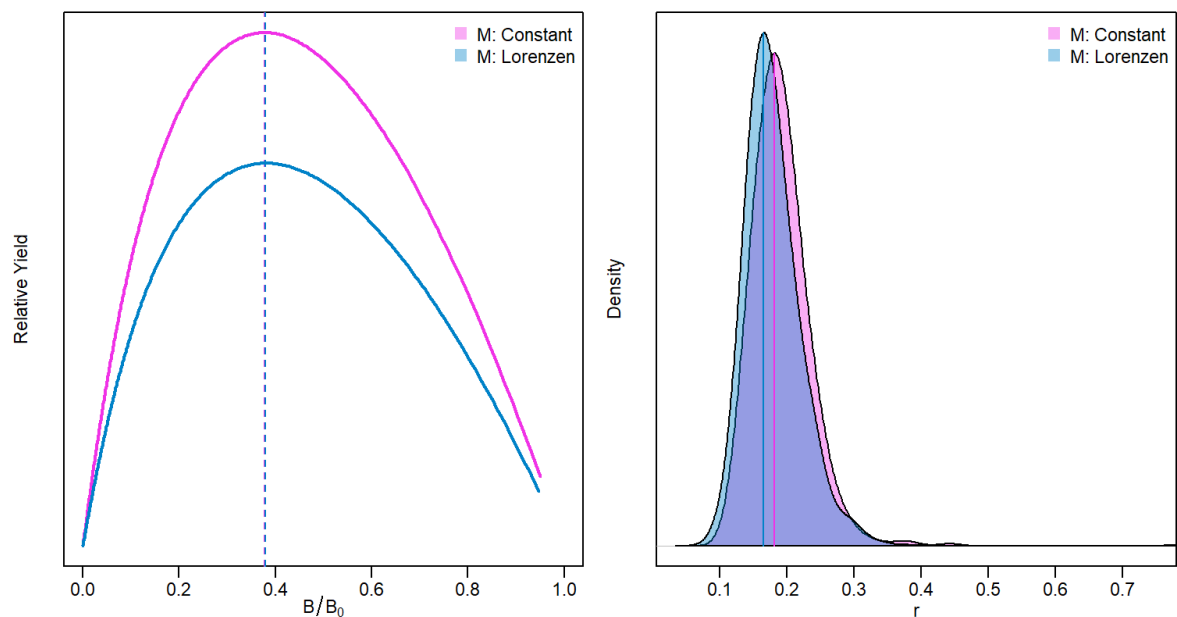


Figure 4. Results of Age-Structured Equilibrium Model (ASEM) with a Beverton-Holt Stock-Recruitment relationship using Monte-Carlo simulations. Left panel: Estimates of yield curves by M ; Right panel: Density distributions of simulated r values by M .

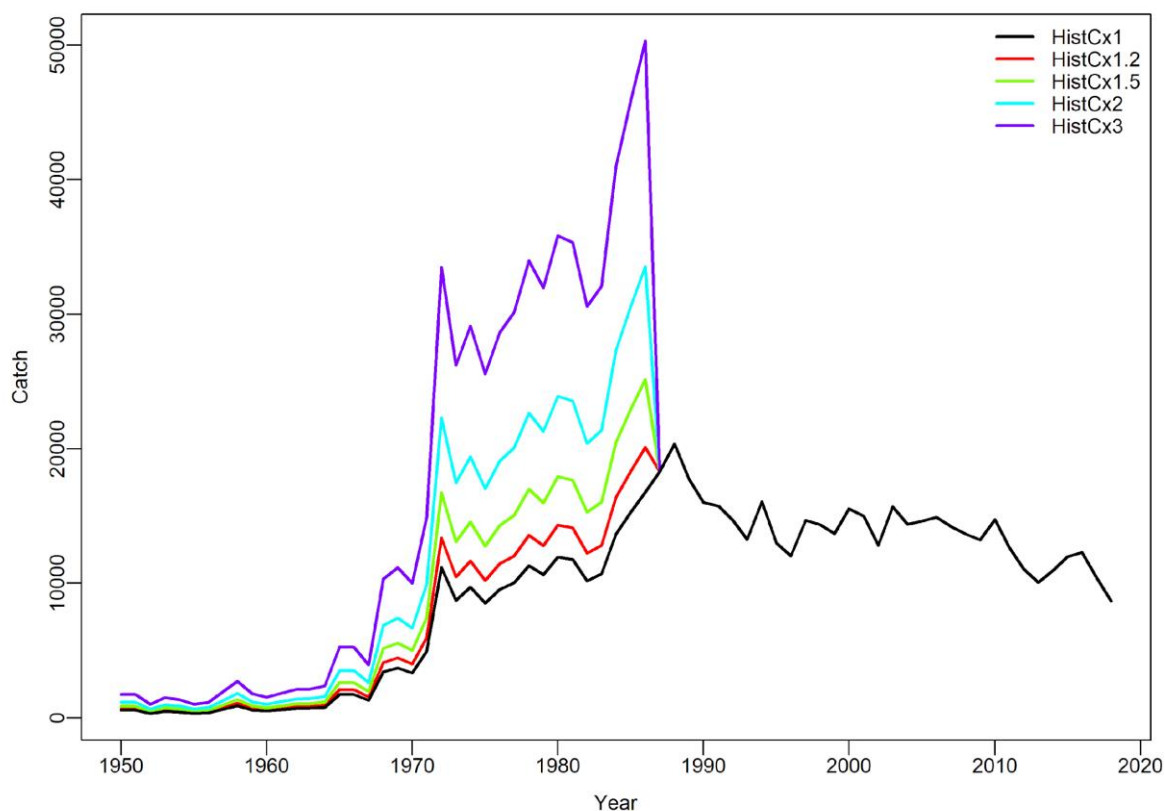


Figure.5 – Alternative catch time series used in sensitivity runs to explore potential impacts of underreporting of early catches 1950-1986 by 20% (HistCx1.2), 50% (HistCx1.5), 100% (HistCx2) and 200% (HistCx3) on the JABBA assessment outputs.

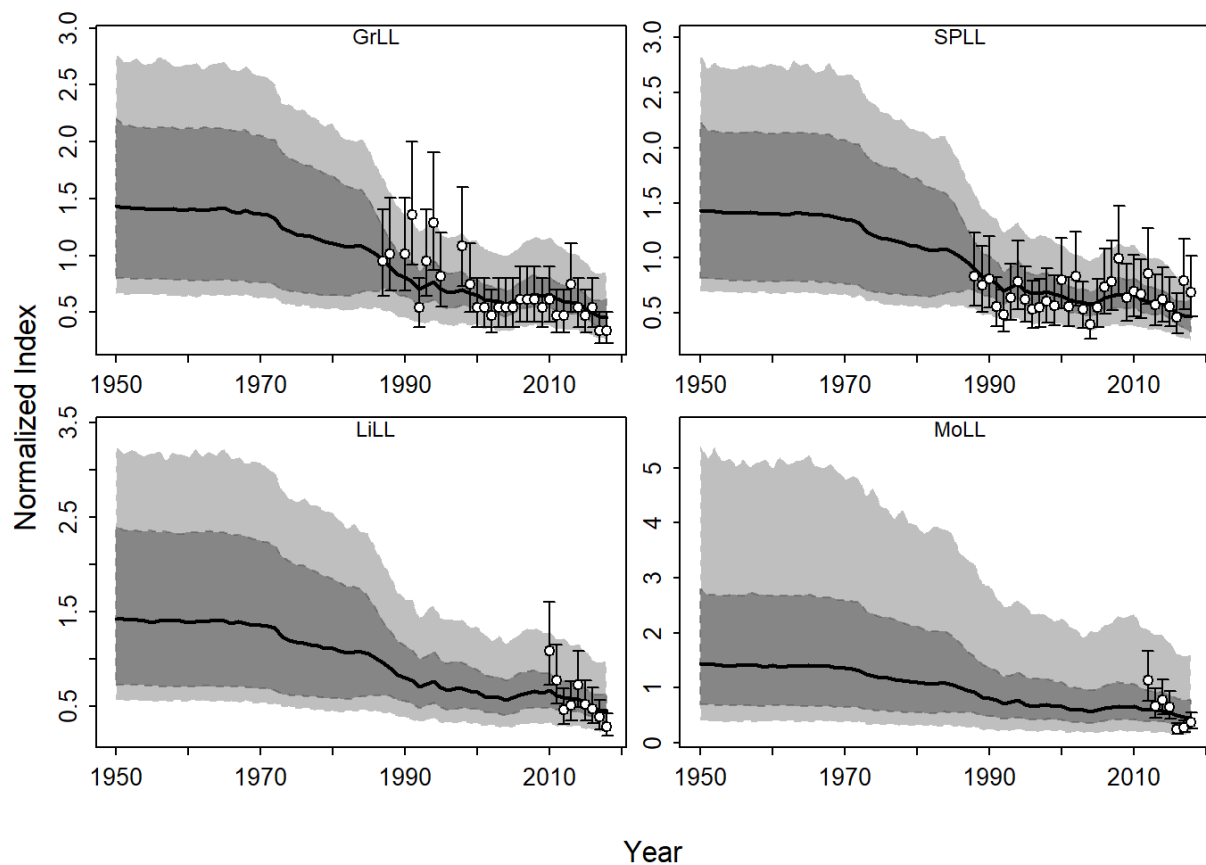


Figure 6. Time-series of observed (circle) with error 95% CIs (error bars) and predicted (solid line) CPUE of Mediterranean swordfish for the Bayesian state-space surplus production model JABBA for the ‘Reference’ scenario. The Dark shaded grey areas show 95% credibility intervals of the expected mean CPUE and light shaded grey area denote the 95% posterior predictive distribution intervals.

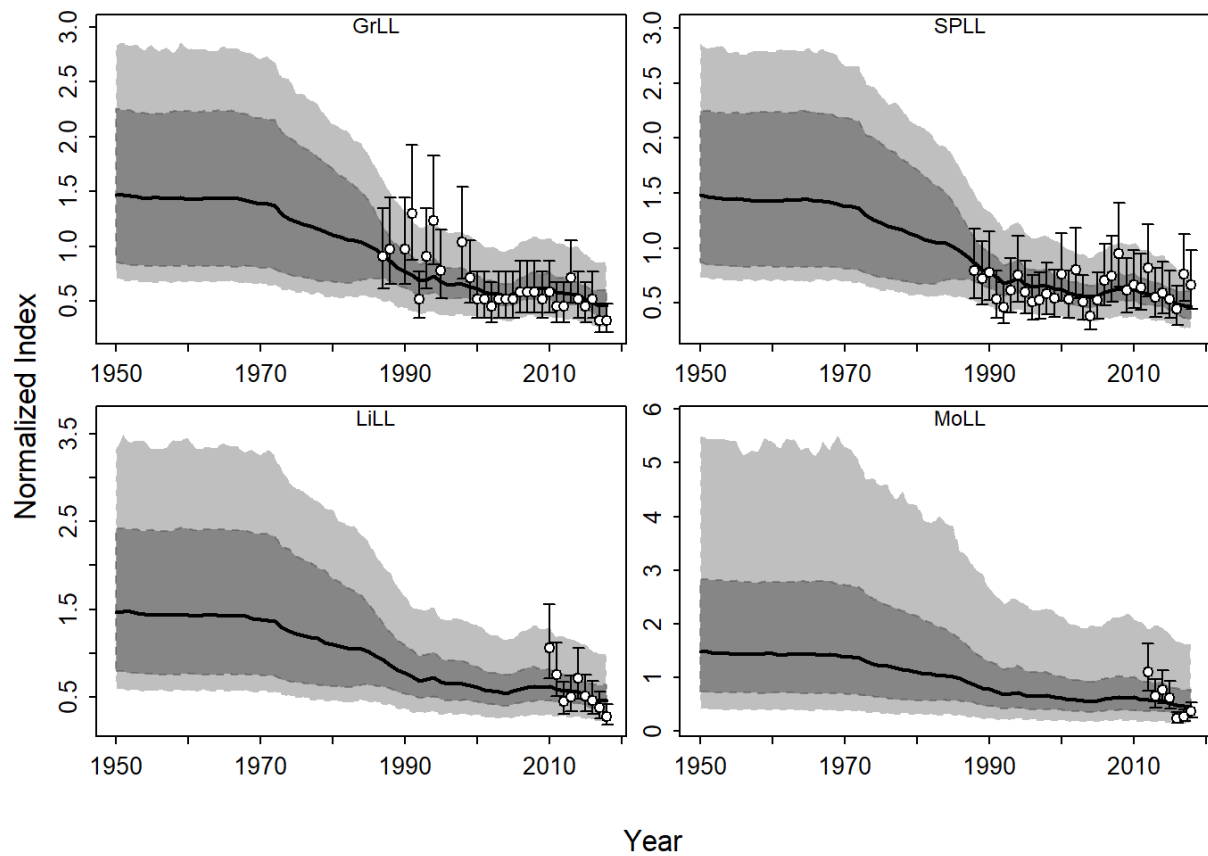


Figure 7. Time-series of observed (circle) with error 95% CIs (error bars) and predicted (solid line) CPUE of Mediterranean swordfish for the Bayesian state-space surplus production model JABBA for the 'ASEM' scenarios. The Dark shaded grey areas show 95% credibility intervals of the expected mean CPUE and light shaded grey area denote the 95% posterior predictive distribution intervals.

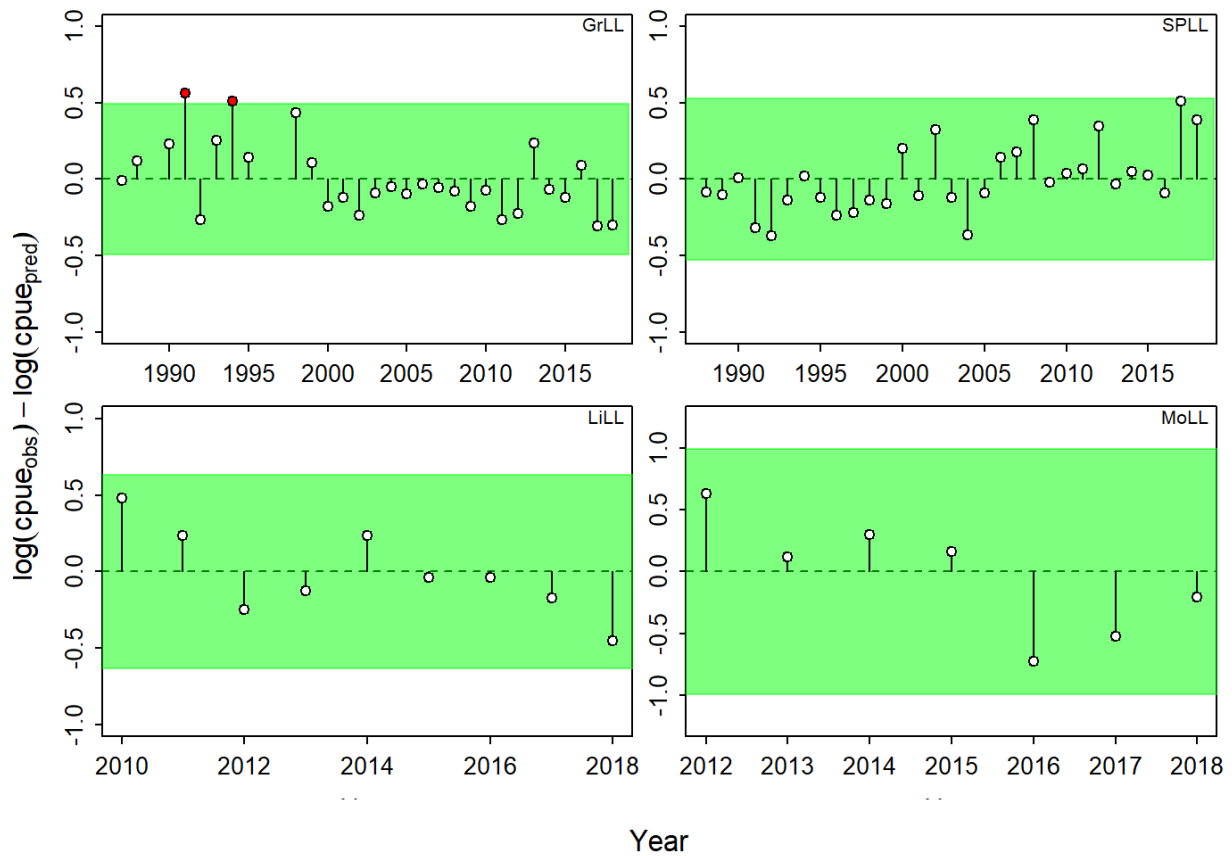


Figure 8. Runs tests to quantitatively evaluate the randomness of the time series of CPUE residuals by fleet for Reference model. Green panels indicate no evidence of lack of randomness of time-series residuals ($p > 0.05$) while red panels (not shown here) indicate the opposite. The inner shaded area shows three standard errors from the overall mean and red circles identify a specific year with residuals greater than this threshold value (3x sigma rule).

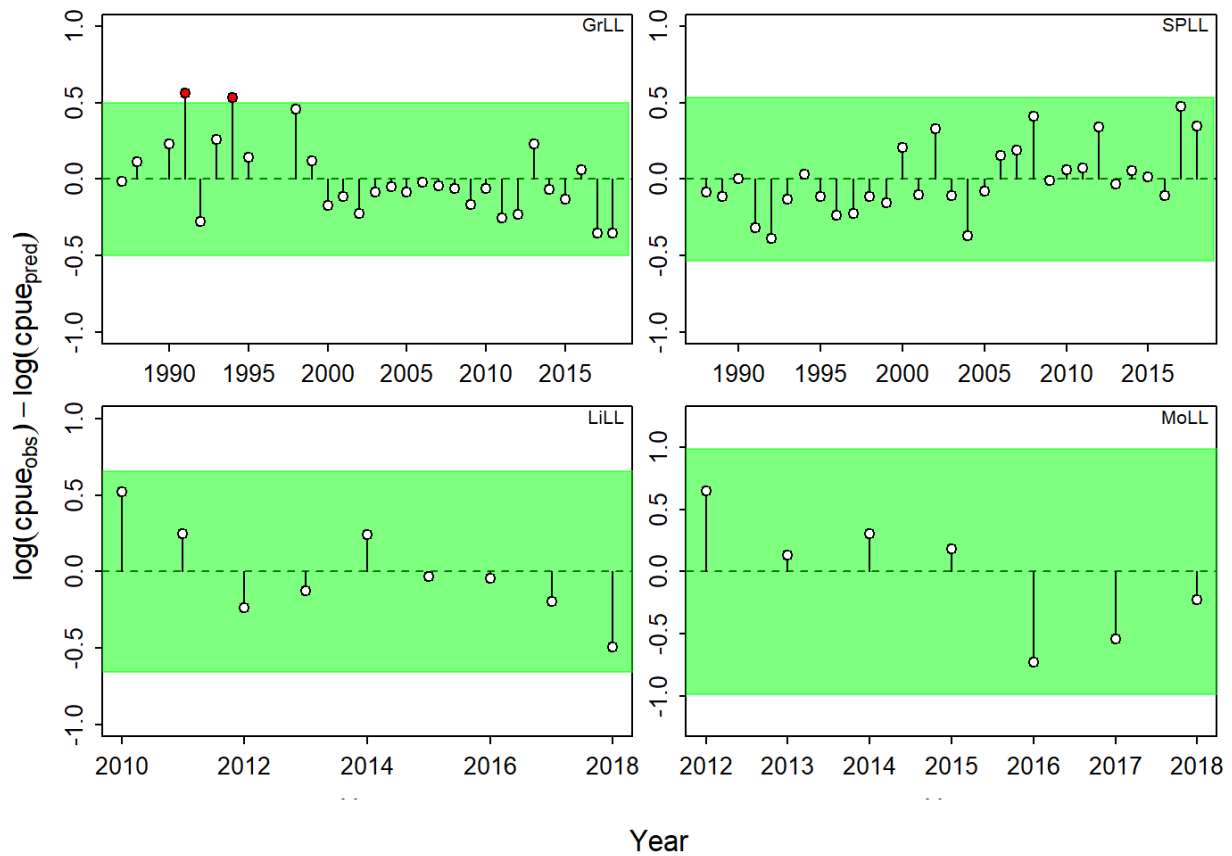


Figure 9. Runs tests to quantitatively evaluate the randomness of the time series of CPUE residuals by fleet for ASEM model. Green panels indicate no evidence of lack of randomness of time-series residuals ($p > 0.05$) while red panels (not shown here) indicate the opposite. The inner shaded area shows three standard errors from the overall mean and red circles identify a specific year with residuals greater than this threshold value (3x sigma rule).

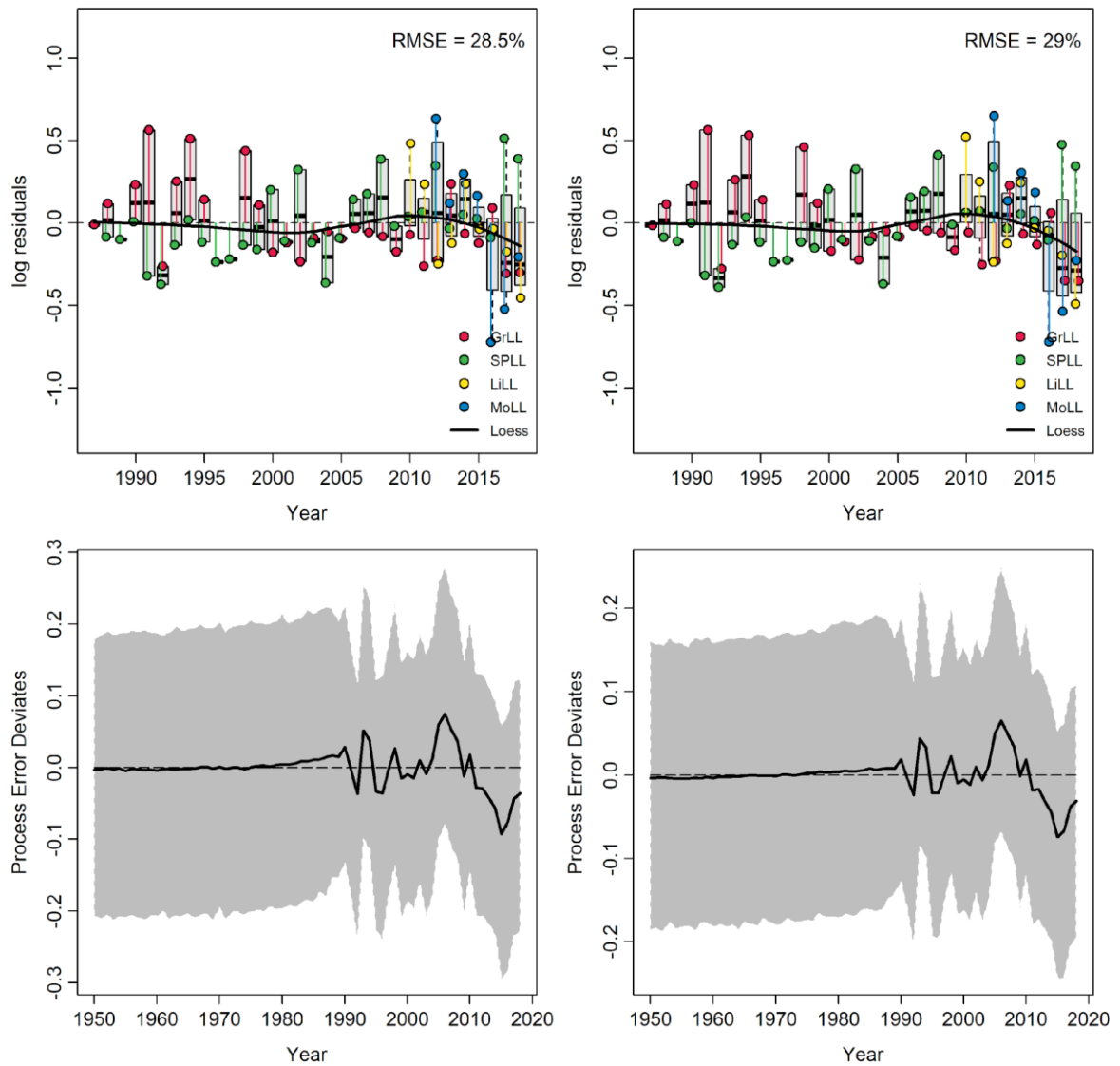


Figure 10. JABBA residual diagnostic plots for alternative sets of CPUE indices examined for each scenario (Left: Reference model; Right: ASEM model) for Mediterranean swordfish. Top panels: Boxplots indicating the median and quantiles of all residuals available for any given year, and solid black lines indicate a loess smoother through all residuals. Bottom panels: Process error deviates (median: solid line) with shaded grey area indicating 95% credibility intervals.

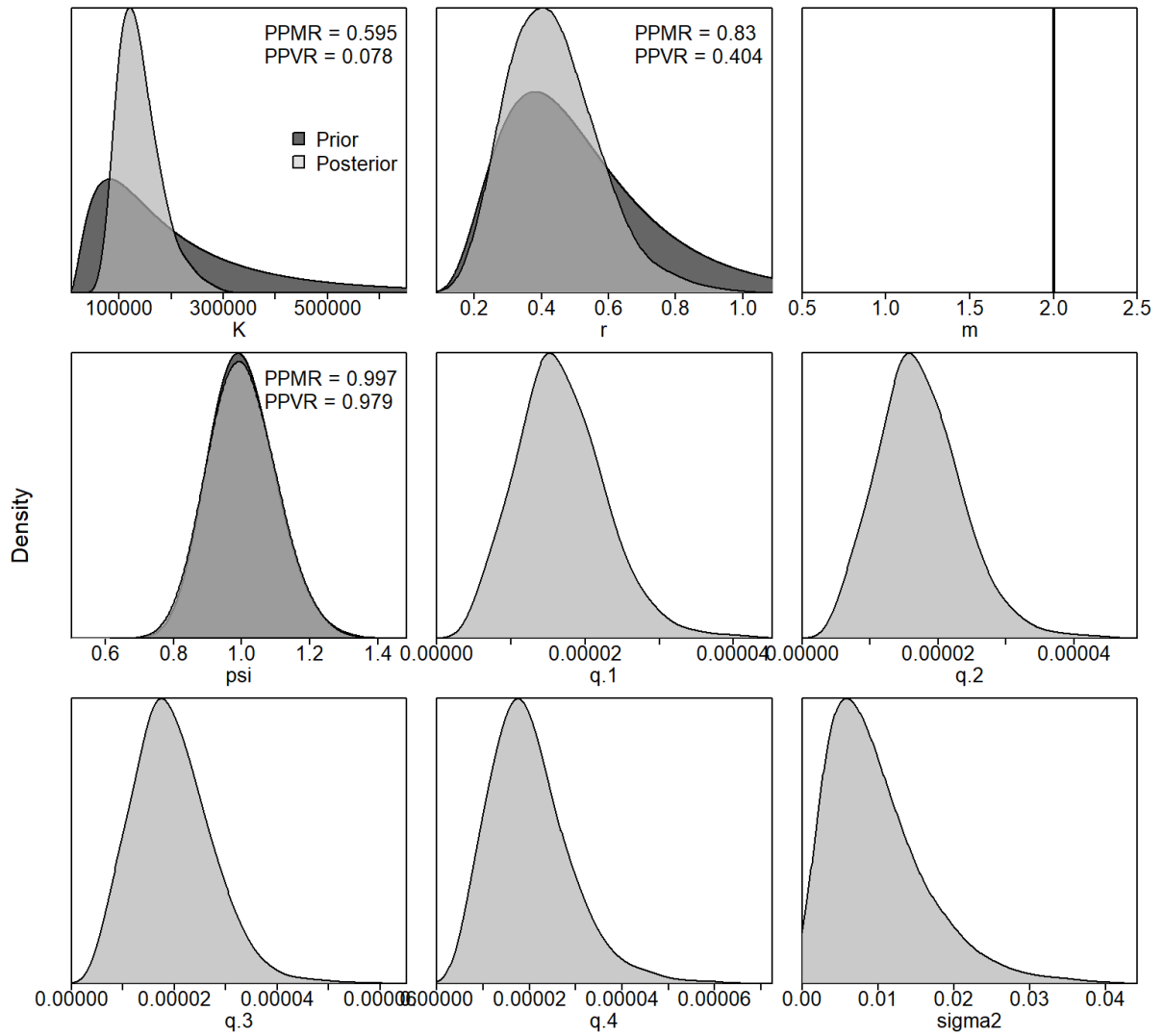


Figure 11. Prior and posterior distributions of various model and management parameters for the Bayesian state-space surplus production model (Reference model) for Mediterranean swordfish. PPMR: Posterior to Prior Ratio of Means; PPVR: Posterior to Prior Ratio of Variances.

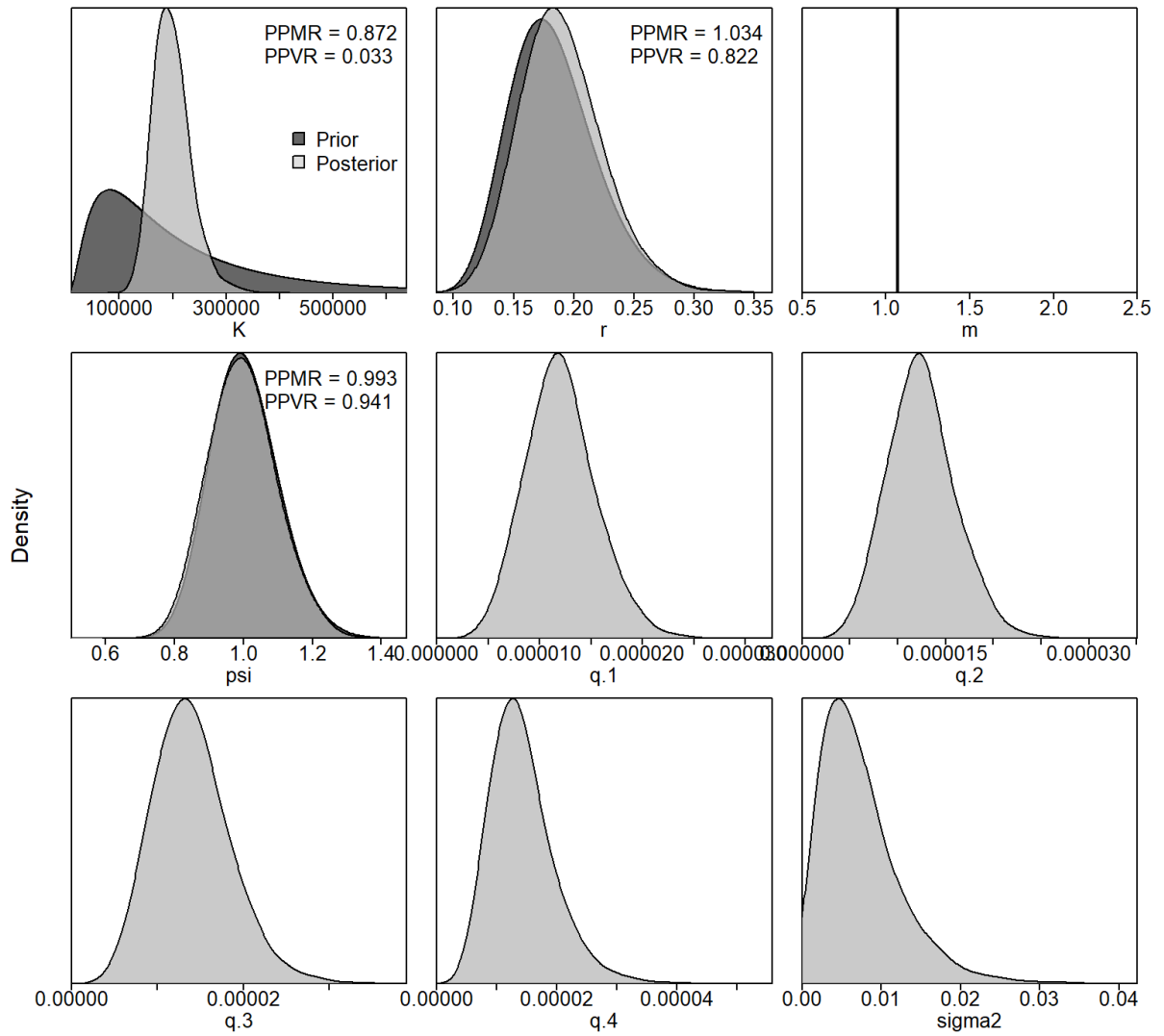


Figure 12. Prior and posterior distributions of various model and management parameters for the Bayesian state-space surplus production model (ASEM model) for Mediterranean swordfish. PPMR: Posterior to Prior Ratio of Means; PPVR: Posterior to Prior Ratio of Variances.

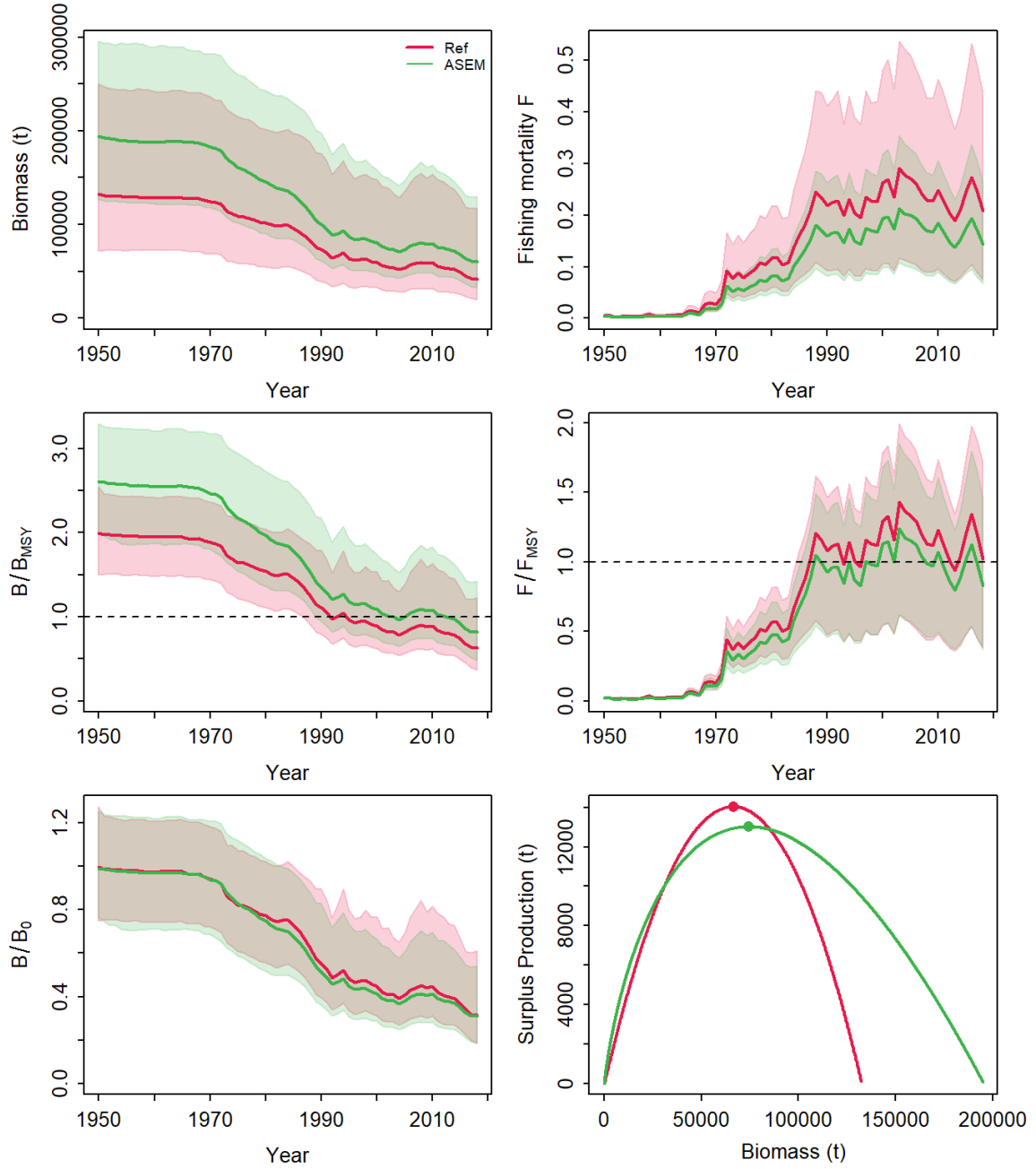


Figure 13. Trends in biomass and fishing mortality (upper panels), biomass relative to B_{MSY} (B/B_{MSY}) and fishing mortality relative to F_{MSY} (F/F_{MSY}) (middle panels) and biomass relative to K (B/B_0) and surplus production curve (bottom panels) for each scenario from the Bayesian state-space surplus production model fits to Mediterranean swordfish.

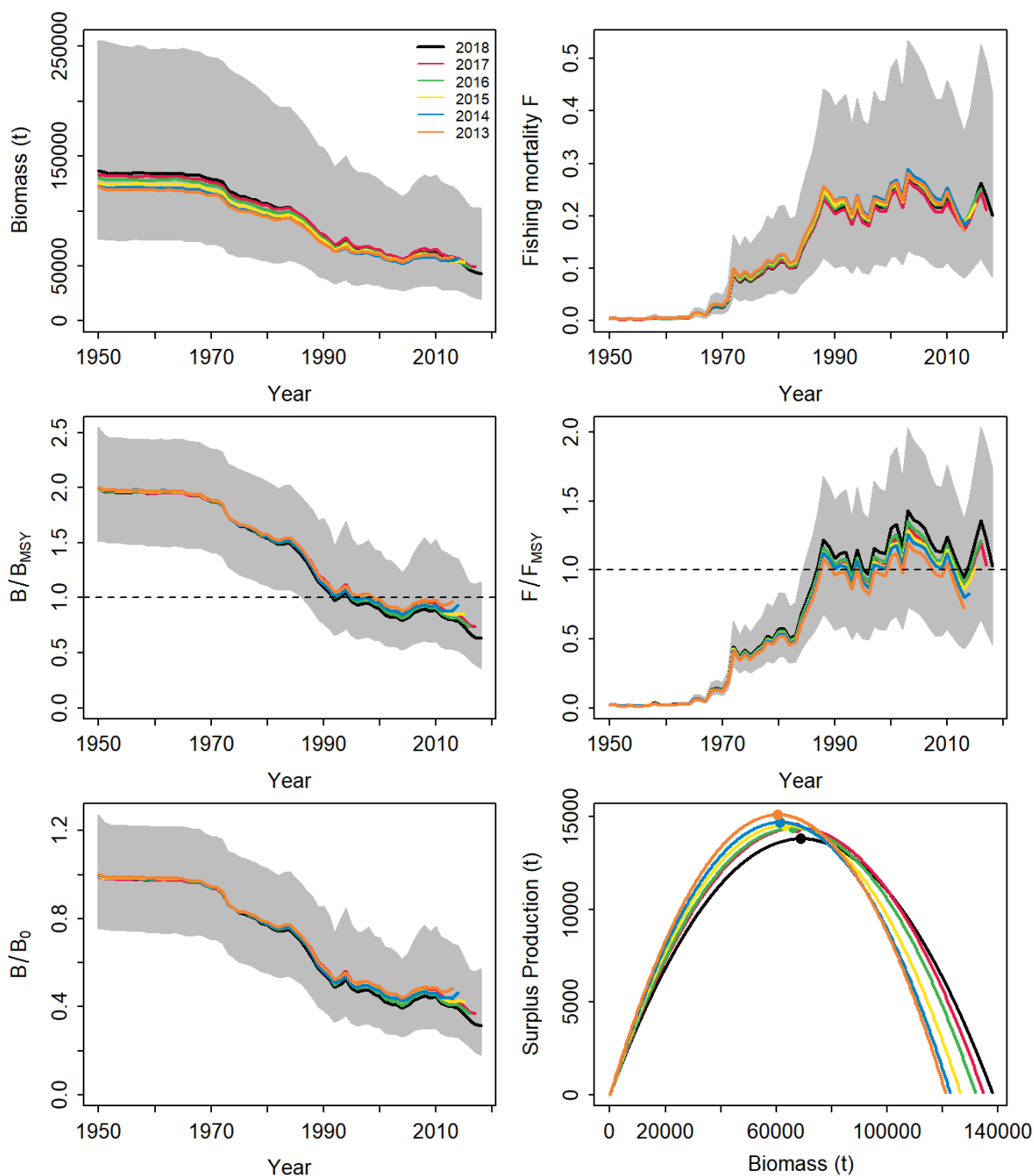


Figure 14. Retrospective analysis performed to the Reference model, by removing one year at a time sequentially ($n=5$) and predicting the trends in biomass and fishing mortality (upper panels), biomass relative to B_{MSY} (B/B_{MSY}) and fishing mortality relative to F_{MSY} (F/F_{MSY}) (middle panels) and biomass relative to K (B/K) and surplus production curve (bottom panels) for each scenario from the Bayesian state-space surplus production model fits to Mediterranean swordfish.

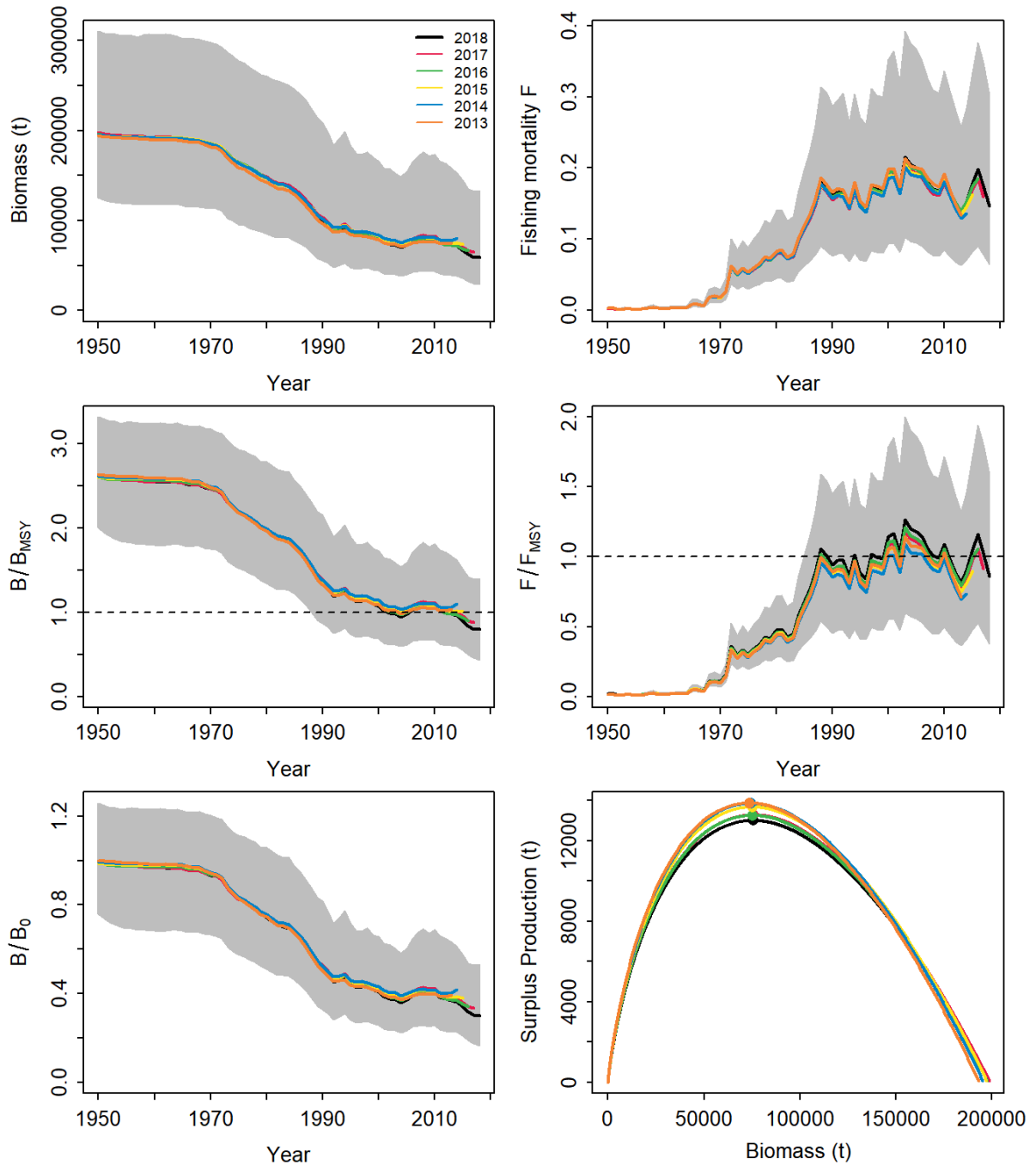


Figure 15. Retrospective analysis performed to the ASEM model, by removing one year at a time sequentially ($n=5$) and predicting the trends in biomass and fishing mortality (upper panels), biomass relative to B_{MSY} (B/B_{MSY}) and fishing mortality relative to F_{MSY} (F/F_{MSY}) (middle panels) and biomass relative to K (B/K) and surplus production curve (bottom panels) for each scenario from the Bayesian state-space surplus production model fits to Mediterranean swordfish.

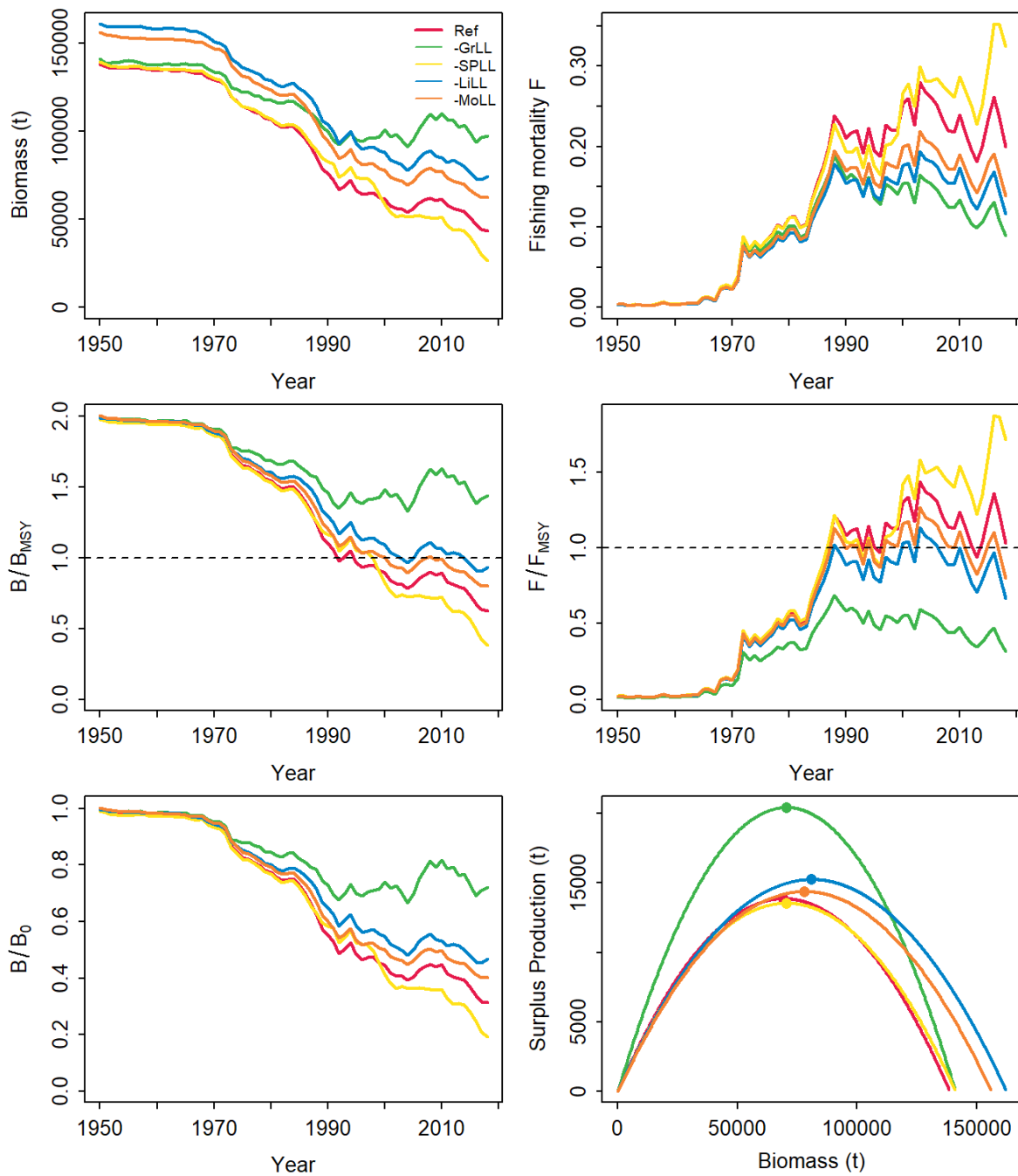


Figure 16. Jackknife index analysis performed to the Reference model, by removing one CPUE fleet at a time and predicting the trends in biomass and fishing mortality (upper panels), biomass relative to B_{MSY} (B/B_{MSY}) and fishing mortality relative to F_{MSY} (F/F_{MSY}) (middle panels) and biomass relative to K (B/K) and surplus production curve (bottom panels) for each scenario from the Bayesian state-space surplus production model fits to Mediterranean swordfish.

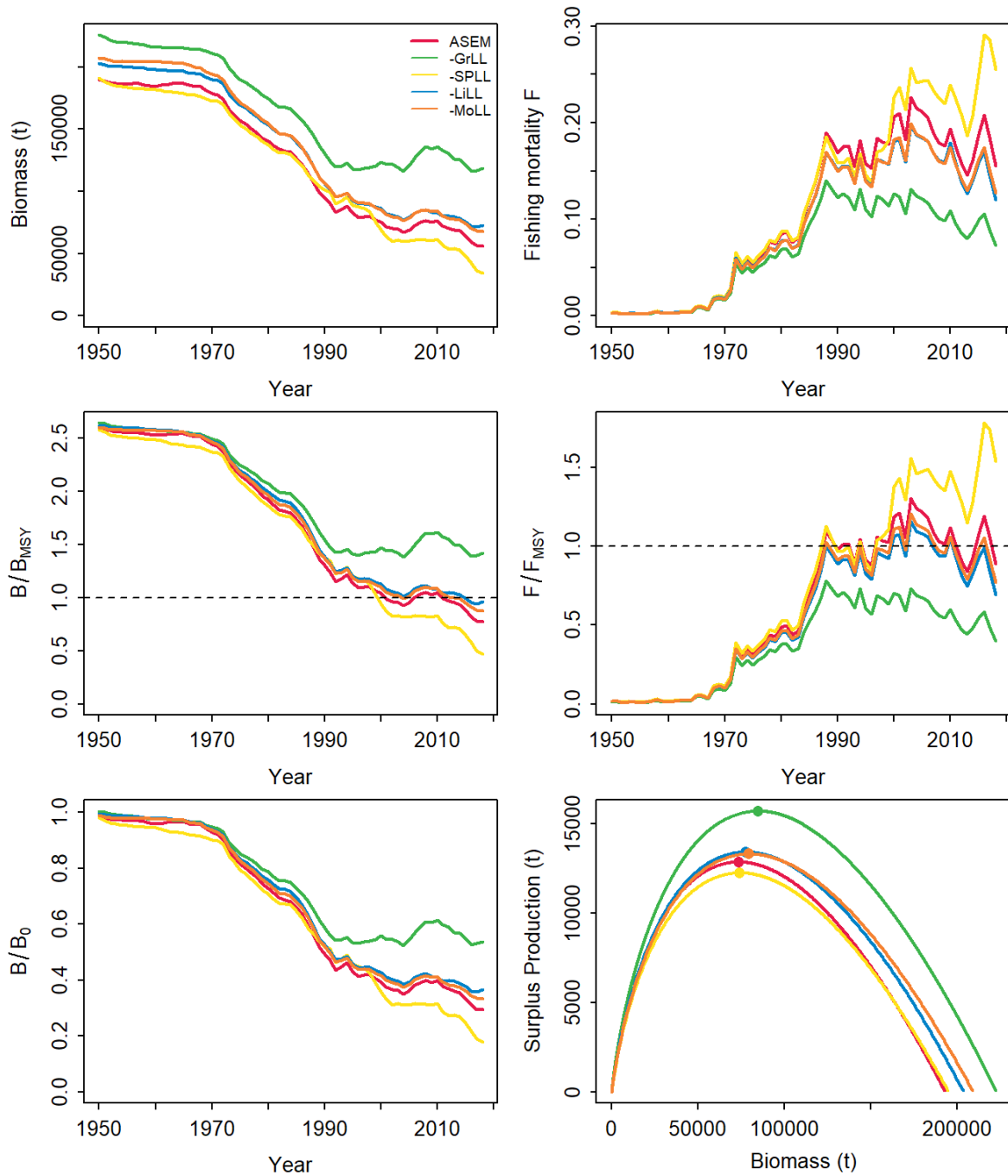


Figure 17. Jackknife index analysis performed to the ASEM model, by removing one CPUE fleet at a time and predicting the trends in biomass and fishing mortality (upper panels), biomass relative to B_{MSY} (B/B_{MSY}) and fishing mortality relative to F_{MSY} (F/F_{MSY}) (middle panels) and biomass relative to K (B/K) and surplus production curve (bottom panels) for each scenario from the Bayesian state-space surplus production model fits to Mediterranean swordfish.

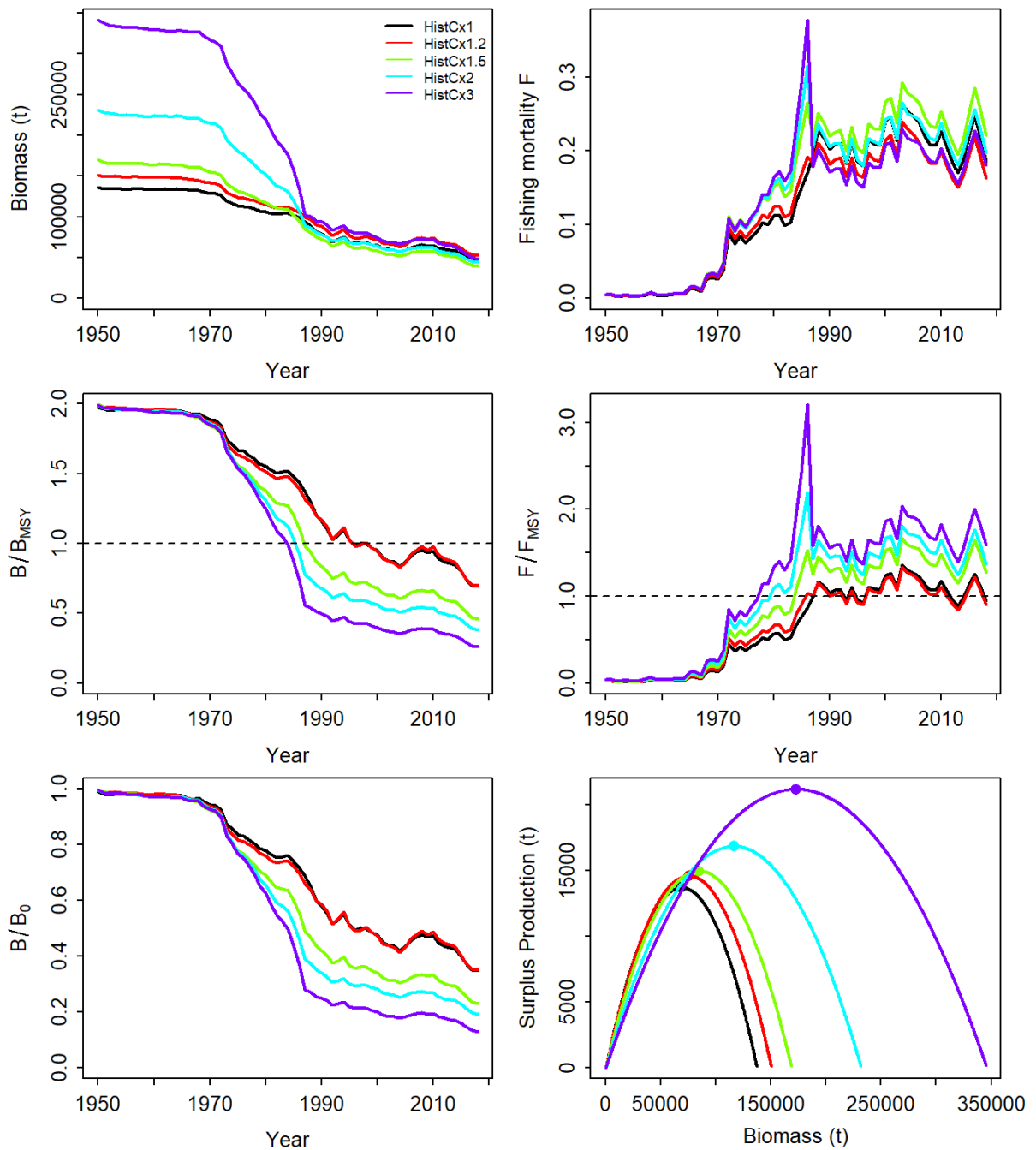


Figure 18. Sensitivity analysis performed to the Reference model for each assumption with regards historical catches (multiplying the early catches from 1950 to 1986 by factors of 1.2, 1.5, 2 and 3) and predicting the trends in biomass and fishing mortality (upper panels), biomass relative to B_{MSY} (B/B_{MSY}) and fishing mortality relative to F_{MSY} (F/F_{MSY}) (middle panels) and biomass relative to K (B/K) and surplus production curve (bottom panels) for each scenario from the Bayesian state-space surplus production model fits to Mediterranean swordfish.

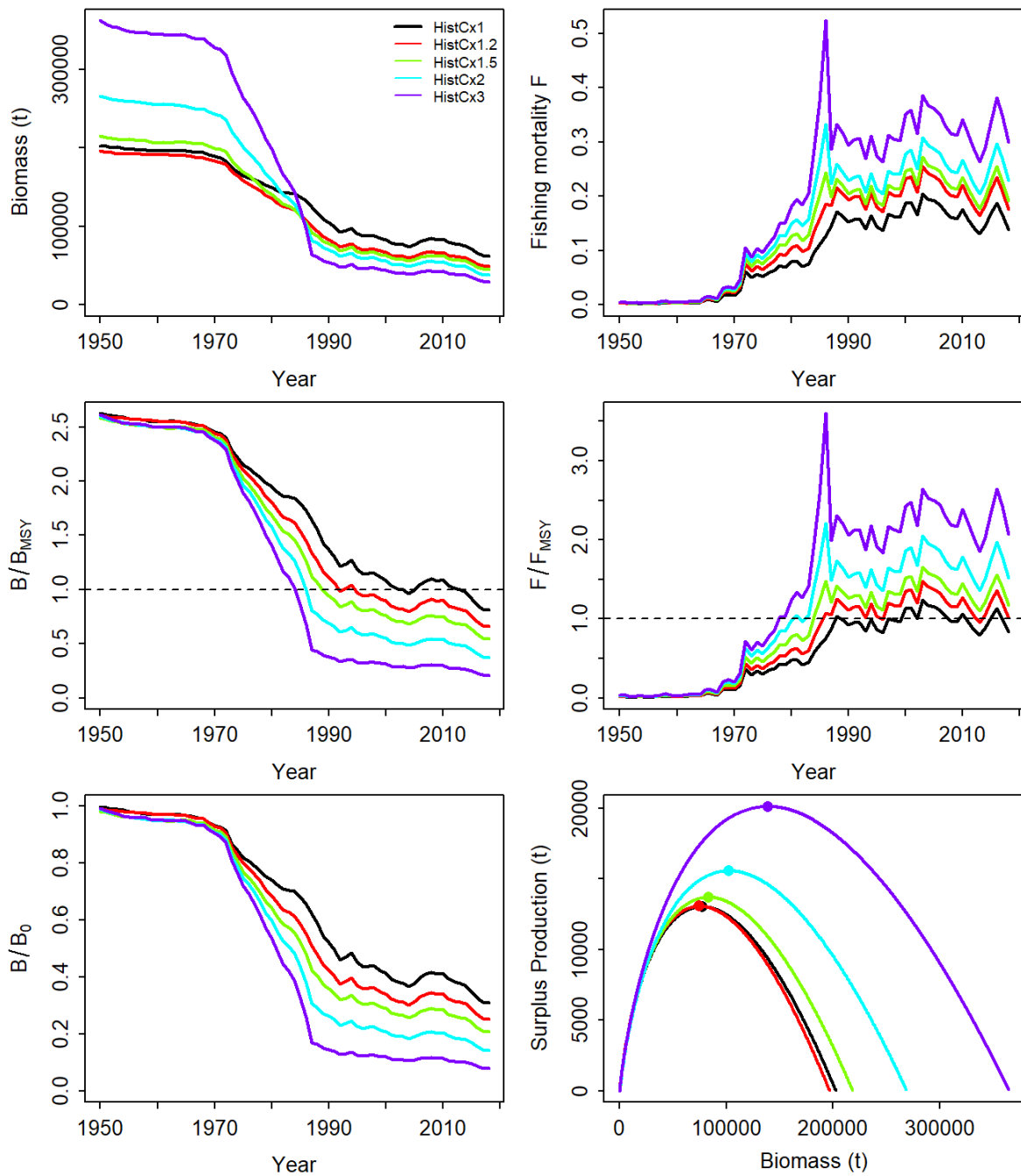


Figure 19. Sensitivity analysis performed to the ASEM model for each assumption with regards historical catches (multiplying the early catches from 1950 to 1986 by factors of 1.2, 1.5, 2 and 3) and predicting the trends in biomass and fishing mortality (upper panels), biomass relative to B_{MSY} (B/B_{MSY}) and fishing mortality relative to F_{MSY} (F/F_{MSY}) (middle panels) and biomass relative to K (B/K) and surplus production curve (bottom panels) for each scenario from the Bayesian state-space surplus production model fits to Mediterranean swordfish.

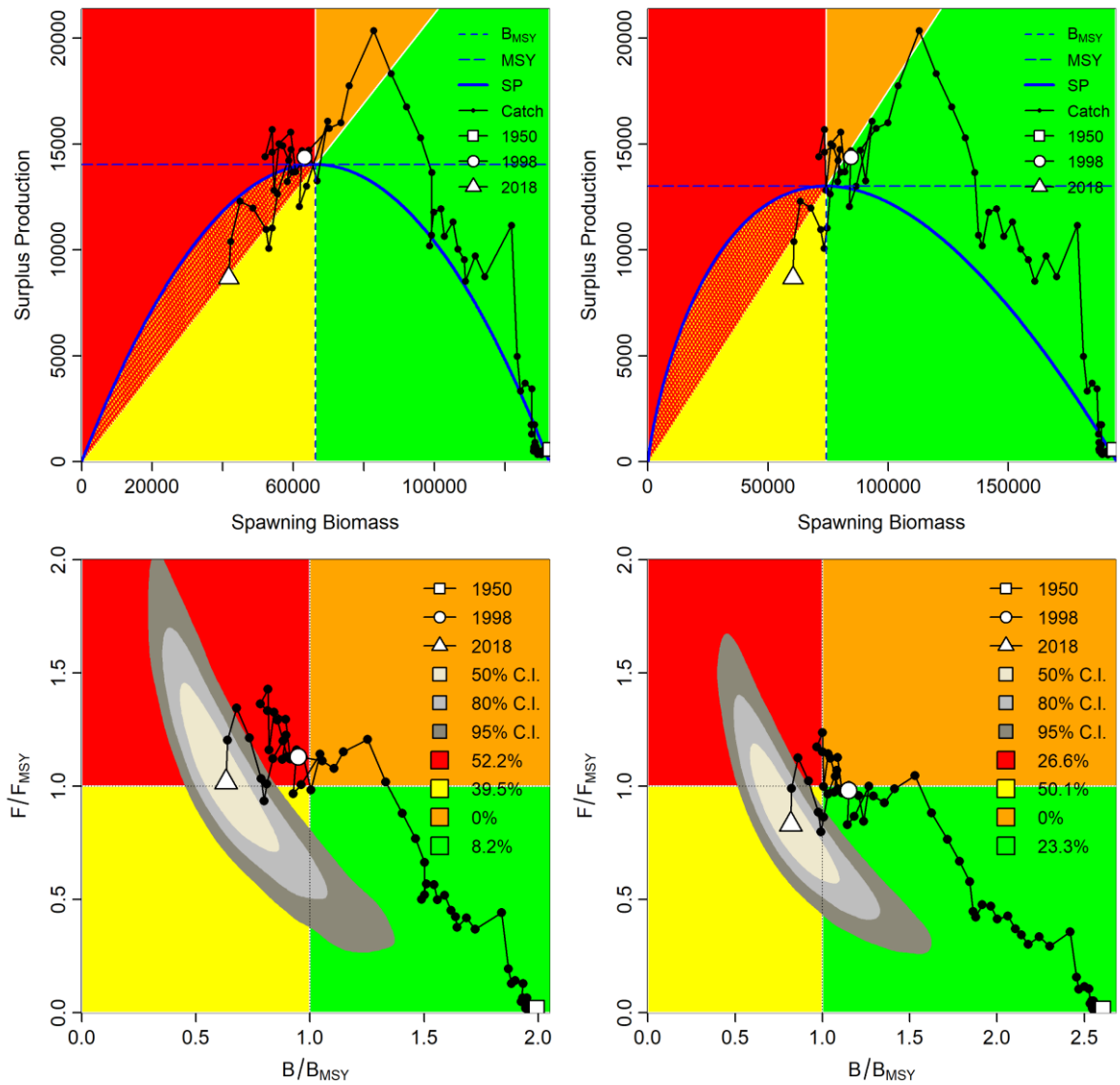


Figure 20. JABBA surplus production phase plot for the Reference Model (left panels) and the ASEM model (right panels) showing trajectories of the catches in relation to B_{MSY} and MSY (top panels) and Kobe phase plot showing estimated trajectories (1959-2018) of B/B_{MSY} and F/F_{MSY} for the Bayesian state-space surplus production model for the Mediterranean swordfish (bottom panels). Different grey shaded areas denote the 50%, 80%, and 95% credibility interval for the terminal assessment year. The probability of terminal year points falling within each quadrant is indicated in the figure legend.